Actuarial Non-Life Insurance

Giuseppe d'Aprile, Philip Salme

A.Y. 2022-2023

Contents

Chapter 1

Introduction

The general Framework for Non-Life Underwriting Profits and Losses (PL_t) of year t for a specific line of business (LoB) is defined as:

$$PL_t = B_t - X_t' - V_t^S + V_{t-1}^S - E_t =$$

$$= \underbrace{(B_t' - V_t^P + V_{t-1}^P) - X_t^{'CY} - V_t^{S,CY} - E_t}_{Pricing\ Profit} + \underbrace{V_{t-1}^S - X_t^{PY} - V_t^{S,PY}}_{Claim\ Reserve\ Sufficiency}$$

B_t (earned premiums) is decomposed into: what I collected B_t' plus the variation in the reserve premium $(V_t^P + V_{t-1}^P)$;
X_t' (amount paid for claims in the calendar year t) is decomposed into: what I paid for claims of the calendar year $X_t'^{CY}$ and what I paid for claims of the previous years $X_t'^{PY}$;
V_t^S (final claim reserve of the calendar year) is decomposed into: what I reserved for claims occurred in the calendar year ($V_t^{S,CY}$) and what I reserved for claims occurred in previous years ($V_t^{S,PY}$);
E_t the greater part of expense are: acquisition expenses and general expenses. The settlement expenses are not inside E_t but are inside the payments or in the reserve.
In this formula, financial profits, or losses, Reinsurance Profits or losses and aggregation between LoBs is not taken into account.
We can look at this formula from two different point of views:

- Ex-Ante: Interpreting the values involved as random Variables, in order to estimate the future values, or to consider the volatility of the estimates, given a specific method.
- Ex-Post: Measure the real, profit or loss made in the year t, knowing all the components.

And it permits us to:

- Analyzing the distribution of the different Random Variables involved (for capital requirement)

- measure the origin of the profit/losses (Premium and Reserve Profit or Losses);

Different Valuation Criteria can lead to different valuation of PL_t, such as GAAP, IFRS-17, Solvency II. The decomposition of the P&L can be done independently from the type of BS we are using.
In the Local Accounting principle we have:
Premium Reserve:
It is composed by:

- Unearned Premium Reserve (based on a pro-rata temporis approach);
- Additional unexpired Risk Reserve (eventual extra reserve to cover future obligation).

Outstanding Claims Reserve:
of which the main features are:

- No discounting;
- Gross of Recoveries: in the same Lob we can have some credit to policyholder. For example in MTPL we can have retention (1 million) and the claim has an higher amount (2 million). The law tells the insurance undertaking to pay the amount (2 million) and then to ask the money back to the policyholder.
  This leads to a strange situation: if the undertaking pays the claim today and it receives the money back after 10 years, the company can have negative cell in the upper triangle and it cannot use some models (i.e. a model based on the Log-Normal distribution);
- Case outstanding (riserva di inventario): evaluated claim by claim (case by case). Made by the claim adjusters (liquidatori).
  Pro: availability of historical data
  Cons: subjectivity of the claim adjusters;
- Provisions for future development on known (reported) claims (RBNS);
- Claim adjustment expenses: expenses for evaluating the claim. They are allocated (dirette) and unallocated expenses (indirette) (Allocated loss adjustment expenses + Unallocated loss adjustment expenses ALAE+ULAE) They can go in payments or in reserve;
- Provision for claims incurred but not reported (IBNR): this are not included in the case reserve (since it is made up claim by claim, so we have the information only when the claim is reported).
  It can be present in our actuarial model;
- Adjustments (changing claims environment).

Chapter 2

Deterministic Methods

2.1 Notation and Development Triangles

Reserve can be based on:

- Claims specific factors (number of claims, severity, ...)
- External factors (weather condition, natural catastrophe, ...)

The basic Tools used for the analysis are the so-called Triangles

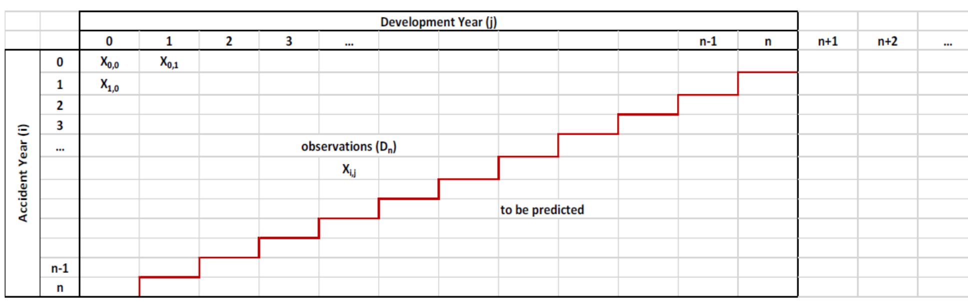

Figure 2.1: The generic development triangle

In the rows we have the Accident Year or Reporting Year (AY or RY), the Underwriting year is used by the reinsurance company, indexed by i, while on the columns we have the Development Year (DY), indexed by j.

- Accident year: it is the most important type of classification, we attribute a claim to that row it is occurred in that year. The main advantage of this classification is that the company knows all the claims occurred in a specific year, so it has a pool of homogeneous contracts occurred in the same year. A drawback is that it omits the IBNR claims;

- Reporting year: the claim goes in that cell if it is reported that year (independently by the year of occurrence of the claim). The advantage is that the company already knows the total number of claims for every cell (since there is not the presence of IBNR claims);
- Underwriting year: the claim is attributed to a year if it refers to a policy underwritten in that year (there is a direct relation between claim and policy). This is done because reinsurance treaties are mono-annual contracts.

Usually both i and j start from 0 up to n, but variations of the triangles are possible with $i > j$ (Trapezoid) or with $j > i$ (Tail Triangles).
The Calendar Years are obtained as the diagonals of the triangles, so that $i + j = k$ gives the k-th Calendar Year.
Some of the most used Triangles in actuarial practice are:

- Incremental Claims Paid $P_{i,j}$, or Cumulative Claims Paid $C_{i,j} = \sum_{h=0}^{j} P_{i,j}$
- Incremental amount at Reserve (Case Reserve or Booked Reserve, Best Estimate) $R_{i,j}$
- Number of Reported Claims by accident year $d_{i,j}$, or aggregated by Report Year d_i
- Number of Paid Claims $n_{i,j}$
- Number of Open Claims $r_{i,j}$
- Reopened Claims amount $re_{i,j}$
- Number of Claims Closed without Payment $s_{i,j}$

2.1.1 The Development Triangle as Diagnostic Tool

We can have some particular cases in the actuarial practice, one of these can be the presence of partial payment: the Regulator states that the amount of the payment is considered ($P_{i,j}$) while the number ($n_{i,j}$) is not. Following this method we can have biased result regarding the average paid cost ($m_{i,j} = \frac{P_{i,j}}{n_{i,j}}$) since we will have an increase in the numerator but the denominator will remain the same. This biased computation does not occur for the Reserve, where partial payment is considered both in the amount of the reserve and in the number of open claims (in the reserve): in fact, even if the insurance company has a partial payment and it uses a part of the reserve to cover it, that outlay will affect both $R_{i,j}$ (equal to the partial amount) and $r_{i,j}$ (counting as 1). Remember that while $r_{i,j}$ can be computed mathematically (there is a specific formula), the amount of the reserve is an estimation made by the insurer.
We can build another type of triangle $s_{i,j} - re_{i,j}$ that can have positive or negative values. Usually, in the first columns positive quantities are present

(since the amount of closed claims w/o payment is higher than the reopen ones), while after some years (column 5-6-7) negative values could be present (because is more probable to reopen a closed claim w/o payment rather than to close an open claim).

The Settlement Speed

A very important indicator in the Non-Life practise is the settlement speed: it tells the insurer the amount or the number of claims **paid** with respect to the total number of claims **occurred**.
As just said, the settlement speed can be computed based on the amount or the number (of claims) and can be considered for cumulative or incremental values. We will focus more on the settlement speed based on cumulative values (both for number and amount).

- Amount (cumulative): $v^A_{i,j} = \dfrac{C_{i,j}}{C_{i,j} + R_{i,j}}$
- Number (cumulative): $v^N_{i,j} = \dfrac{n^c_{i,j}}{n^c_{i,j} + r_{i,j}}$

The maximum value of the settlement speed is 1 (meaning that the insurer has paid all the claims occurred). The speed of convergence (how fast the settlement speed reaches 1) can change considering different LoB: faster is the speed of convergence and and less is the insurer affected by the level of inflation, but notice that if the insurer pays all the claim in the year N, it can have some problems in terms of amount paid (paying all the 2022 claims in the 2022 can lead to an overestimation of the cost, since the most important claims are usually paid after different years).

2.2 Paid Chain Ladder Method

It is one of the most used Methods in practice, given its simplicity and the fact that only relies on the information of the upper triangle. The quality of the estimates depends on the Stochastic Model generating the Data.

2.2.1 Assumptions

The use of run-off triangles for estimating future claims loss payments requires the assumption that the development of settled claims losses follows the same pattern for every claims occurrence year. Furthermore, it is assumed that this pattern will hold for future payments, i.e. the pattern will continue into the future.
It also seems reasonable to expect that the estimates for settlement amounts in the future will be more accurate if all of the available data is used in the estimation. The Chain-Ladder Method is a technique that follows this expectation.

It relies on the use of the cumulative claims loss settlement data for all claims occurrence years rather than considering, for example, only the latest claims occurrence year
Main Assumption is:
All accident years i behave similarly and for cumulative claims we have approximately:

$$C_{i,j+1} = C_{i,j} \cdot \lambda_j \quad \forall \ i, j$$

For given Chain Ladder Factors $\lambda_j > 0$, also called Claim Development Factors. The above assumptions must be verified if possible, and anyway we must take into account that over time loss settlement patterns can change due to:

- Changes in product design and conditions;
- Changes in the claims reporting, assessment and settlement processes;
- Changes in inflation rates;
- Changes in Legal environment
- Abnormally large or small claim settlement amounts.

2.2.2 The Method

The first thing we need to do is to build the triangle of Individual Loss Development Factors, $f_{i,j} = \dfrac{C_{i,j+1}}{C_{i,j}}$. Once the Individual factors have been built, boxplots can be used in order to verify the volatility of the loss development factors for different j. Given the fact that the assumptions require stability in the development factors, the idea is to have the same development factors for the same column, even if this is absurd in real life. The higher is the variability we observe in the factors, the more the assumptions are not satisfied.
In order to build the Development Factors, different aggregation methods are possible, such as arithmetic mean, weighted mean, and variations using only the last 3-4 Calendar years (diagonals), even if this implies that we use less data, it is useful to use it when we have different pattern in the last 3-4 years vs the years before.
The classical Chain-Ladder method estimates the unknown factors λ as:

$$\widehat{\lambda}_j = \frac{\sum_{i=0}^{n-j-1} C_{i,j+1}}{\sum_{i=0}^{n-j-1} C_{i,j}} = \frac{\sum_{i=0}^{n-j-1} f_{i,j} \cdot C_{i,j}}{\sum_{i=0}^{n-j-1} C_{i,j}}$$

When selecting the Claim Development Factors, Actuaries need to verify that:

- Smooth progression of individual age-to-age factors and development factors across DYs;
- Stability of age-to-age factors for the same DY;
- Credibility of the experience;

- Applicability of historical experience;
- Changes in pattern.

By using this method we only have development factors until DY n, if also **Tail Factors** needs to be included, different alternatives are possible:

- Estimation of ultimate Cost of first AY based on the loss reserve;
- Curve Fitting methods (based on the negative exponential behaviour of $\widehat{\lambda}_j$);
- Benchmark Based methods;
- Other Alternatives (Algebraic Methods, bond-type methods, etc.).

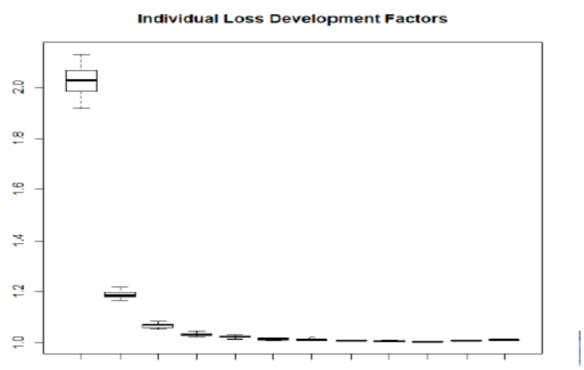

Figure 2.2: Individual Loss Development Factors Box-Plots

All methods allow to estimate $\widehat{\lambda}_j$ for $j > n-1$, since the factors are multiplicative in Chain Ladder methods, we can build $\widehat{\lambda}_\infty = \widehat{\lambda}_n \cdot \widehat{\lambda}_{n+1} \cdot ....$
By using this vector of Development Factors $\widehat{\lambda} = [\widehat{\lambda}_0, \widehat{\lambda}_1, ..., \widehat{\lambda}_{n-1}, \widehat{\lambda}_\infty]$ we rebuild the lower triangle by using recursively the formula:

$$\widehat{C}_{i,j} = C_{i,n-1} \cdot \prod_{h=n-i}^{j-1} \widehat{\lambda}_h$$

for $i = 1, ..., n$ and $j = n-i+1, ..., n$
$\widehat{C}_{i,n}$ represents the ultimate cost for the AY i, while if the tail is present the ultimate cost is: $\widehat{C}_{i,\infty} = \widehat{C}_{i,n} \cdot \widehat{\lambda}_\infty$, $\forall\, i$.
The un-discounted claim reserve by Paid Chain Ladder is:

$$R^{PCL} = \sum_{i=0}^{n} (\widehat{C}_{i,\infty} - C_{i,n-i})$$

2.3 Incurred Chain Ladder

Incurred Chain Ladder method is developed similarly to Paid Chain Ladder, by using the Incurred Losses:

$$I_{i,j} = C_{i,j} + R_{i,j}$$

As for the PCL, the first thing to do is to compute the individual loss development factors, $f^I_{i,j} = \dfrac{I_{i,j+1}}{I_{i,j}}$, and then aggregating the individual ratios we get the final vector of $\widehat{\lambda}^I_j$. The individual development fact now has a different explanation since if the ratio $f_{i,j}$ is lower than 1 the run-off is positive (what we paid in the subsequent year is lower than what we actually reserved), if the ratio is greater than 1 the run-off is negative. The change in the amount to be paid on the next year depends on 2 factors:

- Claims paid in $i, j+1$ can have a different amount than the amount reserved in i, j
- Claims are reserved in i, j can be re-reserved in $i, j+1$ but with a different amount

We have to assume stability in the ratio $f^I_{i,j}$ fixing the column and changing the row (so going down in the triangle), but the elements that affect the volatility of the loss development factor are different in the incurred C-L rather than the paid C-L: if the ratio is lower than one it means that the amount of the reserve in the previous year was higher than the effective amount paid, so the company has been prudent (overestimating the reserve) and this prudence is kept also for others accidents years (the problem is not the presence of under or over estimation of the reserve but the stability of the ratio).
One of the main difference with PCL is the fact that in the last development factor, an estimation of the tail is already included, given by the presence of $R_{i,n}$ (so we do not need $\widehat{\lambda}_\infty$ given that the estimation of the tail is already included in $\widehat{\lambda}_{n-1}$).
We could have some problems when we do not have at disposal $R_{0,n}$ given that, for example, we are an internal actuary and the value of the balance sheet reserve is available only at the end of the year. With this method we cannot fit a curve for $\widehat{\lambda}_\infty$ (as we have done with the PCL) because we do not have an asymptotic behavior of the individual link ratios. The Reserve used for obtaining the Incurred Losses can be the Booked Reserve, or the Case Reserve.

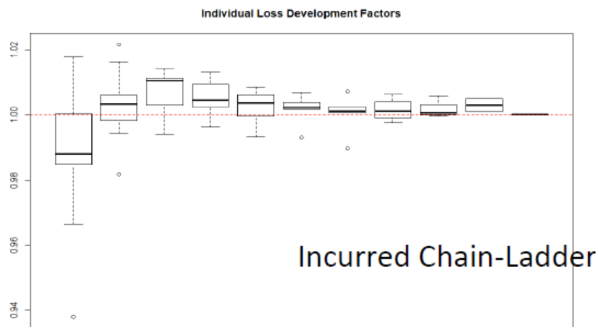

Figure 2.3: Incurred Chain Ladder Individual Factors Box-Plots

In order to estimate the values in the lower triangle we must use the following:

$$\widehat{I}_{i,j} = I_{i,n-i} \cdot \prod_{h=n-i}^{j-1} \widehat{\lambda}^{I}_{i,j}$$

And so the undiscounted Incurred Chain Ladder Reserve is given by:

$$R^{ICL} = \sum_{i=0}^{n} (\widehat{I}_{i,n} - C_{i,n-i})$$

2.3.1 Practical comments on the Chain-Ladder Methodology

Everything said until now does not take into consideration the discounting effect (given that we are still dealing with deterministic method and so with Local GAAP). An important point when we will move to the market consistent evaluation is the presence of discounted reserve.
The main items for discounting are:

- **Incremental payment**: since the risk-free rate term structure is constructed for all the maturities, it is a a curve of spot rate (and not a flat line), so we cannot apply it to a vector of cumulative payment;

- **Reserve used**: when dealing with the Incurred Chain-Ladder we cannot go back to the incremental payment (given that the difference $I_{i,j} - I_{i,j+1}$ returns the run-off, not the incremental payment) and to do that we can use a trick applying the settlement speed constructed as in the Fisher-Lange to the reserve in order to have an estimate of the incremental payments. The problem stand in the reserve used, given that if we use the balance sheet reserve (and we need to have it at our disposal) we are considering also the IBNRs, while if we use the case reserve we do not consider them

2.4 Expected Loss Ratio

Insurance Companies use expected claims technique when entering new lines of business or new geographical territories. Many actuaries use also this method for estimating unpaid claims in most immature periods.
Expected Loss Ratio is a critical component of several other methods, such as Bornhuetter Ferguson and Cape Cod.
With this methodology one can estimate the a-priori, or initial estimate, without using experience observed in the triangle.
The aim of this class of methods is to:

- Quantify the exposure for each accident year;
- Quantify the expected ultimate loss ratio for each accident year.

For estimating the a-priori ultimate loss ratio, the main approaches are based on the estimation used in ratemaking (pricing) or budget purposes. Other times also external data can be useful for computing a-priori ultimate loss ratio (using Market Data).

2.5 Bornhuetter-Ferguson

Bornhuetter-Ferguson method is a mix between development techniques and expected claims techniques. Indeed it can be seen as a credibility weighted mean of two methods:

- In less mature development years, more credibility, and so more weight, is given to the Expected Claims Method;
- In More mature years more weight is given to the Development Methods.

The Main Assumptions Of this Method are:
Unpaid Claims will develop based on expected claims technique
Payment Patterns are the same as those selected in CL method.
Paid **BF** assumes that the ultimate cost U_i^{BF} is obtained as follows:

$$U_i^{BF} = \widehat{C}_{i,\infty}^{PCL} \cdot z_i + \widehat{\mu}_i \cdot (1 - z_i)$$

$z_i = \dfrac{1}{\widehat{CDF}_{n-i}}$ is the credibility factor, determining the weight of the two methodologies. $CDF_j = \displaystyle\prod_{h=j}^{n-1} \widehat{\lambda}_h$. $\mathbf{z_i}$ is a sort of **cumulative claim settlement speed**.
While $\widehat{C}_{i,\infty}^{PCL}$ is the Paid Chain Ladder Ultimate Cost estimation.

$$U_i^{BF} = \widehat{C}_{i,\infty}^{PCL} \cdot \left(\frac{1}{\widehat{CDF}_{n-i}} \right) + \widehat{\mu}_i \cdot \left(1 - \frac{1}{\widehat{CDF}_{n-i}} \right)$$

$$\mathbf{C^{PCL}_{i,n-i}} + \widehat{\mu}_i \cdot \left(1 - \frac{1}{\widehat{CDF}_{n-i}}\right)$$

And so the Reserve as usual defined as $U_i - C_{i,n-i}$ is defined as:

$$\mathbf{R^{BF}_i} = \widehat{\mu}_\mathbf{i} \cdot \left(\mathbf{1} - \frac{\mathbf{1}}{\widehat{\mathbf{CDF}}_{\mathbf{n-i}}}\right)$$

It is easy to compare to the PCL reserve:

$$R^{PCL}_i = \widehat{C}^{PCL}_{i,\infty} \cdot \left(1 - \frac{1}{\widehat{CDF}_{n-i}}\right)$$

Comments on the CDF

The Cumulative development factor is important for 2 reasons:

- CDF used for estimating the ultimate cost in a faster way:

$$\widehat{C}_{i,\infty} = C_{i,n-i} \cdot CDF_{n-i}$$

- It can be seen as a sort of cumulative settlement speed:

2.5.1 Incurred BF

THe concept of Paid BF can also be extended to Incurred Claims, in this case we would have that the ultimate cost would be:

$$U^{BF}_i = C^{ICL}_{i,n-i} \cdot z_i + \widehat{\mu}_i \cdot (1 - z_i)$$

In this case, given the fact that $\widehat{\lambda}^I_h$ are usually very close to 1, and so we could have values of z_i grater than 1.

2.6 Cape Cod

In the Cape Cod Variant of the Expected Claims methodology we have:

$$U^{CC}_i = C_{i,n-i} + B_i \cdot \frac{\sum_{i=0}^n C_{i,n-i}}{\sum_{i=0}^n B_i \cdot z_i} \cdot (1 - z_i)$$

where B_i represent the earned premiums.
Under PCL patterns we have:

$$U^{CC}_i = C_{i,n-i} + B_i \cdot \frac{\sum_{i=0}^n \frac{C_{i,n-i}}{B_i \cdot z_i}}{\sum_{i=0}^n B_i \cdot z_i} \cdot (1 - z_i) = C_{i,n-i} + B_i \cdot \frac{\sum_{i=0}^n ULR^{PCL}_i \cdot B_i \cdot z_i}{\sum_{i=0}^n B_i \cdot z_i} \cdot (1 - z_i)$$

2.7 Fisher-Lange

This methodology was proposed in 1971. It is an average cost method, where outstanding claims are obtained by deriving a separate estimation of frequency and severity.
In this methodology we use the following:

$$P_{i,j} = n_{i,j} \cdot m_{i,j}$$

- $n_{i,j}$ is the **incremental** number of claims, which is supposed to be related to the number of open claimspaid, $r_{i,j}$ through two specific coefficients: **percentage of open claims** having a payment in the future, and **claim settlement speed** representing the timing of the claims count payments estimated to have effectively a payment in the future.

- $m_{i,j}$ is estimated on the basis of the observed triangles and of the assumption on future inflation rates.

The objects to be estimated are 4:

- $aliq_{i,j}$ is the proportion of open claims having effectively a payment in the future, regarding implicitly either nil, reopened, and IBNR.

  $$aliq_{i,j} = \frac{\sum_{h=j+1}^{n-i} n_{i,h} + r_{i,n-i}}{r_{i,j}}$$

  From this triangle of value we want to obtain only a vector of size n, one for each column, $\alpha = [\alpha_0, \alpha_1, ..., \alpha_{n-1}]$, in order to obtain this vector, usually we use the arithmetic mean by column of $aliq_{i,j}$, or alternatively some other function (like weighted average, mean of last k values, and so on).

- $v_j^{(n)}$ is the claims settlement speed, it is used to allocate the total amount of future claims to be paid in the different years.

  $$v_j^{(n)} = \frac{n_{n-j,j} \cdot \frac{d_n}{d_{n-j}}}{\sum_{j=1}^{n} n_{n-j,j} \cdot \frac{d_n}{d_{n-j}}}$$

  This is valid for last accident year $i = n$, if we need it for $i = n-1$ then the summation will go from $j = 2$ to n, and so on. d_i is the number of reported claims for Accident Year i.
  If a tail estimation is needed then also the factor $v_{n^+}^{(n)}$, then the summation need to be extended to n^+

- $\widehat{n}_{i,j}$ is the expected number of future payments, for $i + j > n$ and it is computed as:

  $$\widehat{n}_{i,j} = r_{i,n-i} \cdot \alpha_{n-i} \cdot v_j^i$$

- $\widehat{m}_{i,j}$ the average paid cost is estimated for each cell of the future triangle, following this steps:

  1. we select for each DY, the Average cost, which can be the last observed value or a function of the observed values (like for example the mean), by column.
  2. we select the vector of total claim inflation rates (both endogenous and exogenous) ir_h, expected in the future years.
  3. We inflate the chosen average cost of each column, according to the total claim inflation rates:

  $$\widehat{m}_{i,j} = m_{n-j,j} \cdot \prod_{h=n+1}^{i+j} (1 + ir_h)$$

     In this case the value used is the last available average cost.

Once all the latter values have been estimated, we will obtain $\widehat{P}_{i,j} = \widehat{n}_{i,j} \cdot \widehat{m}_{i,j}$, and then as usual the reserve can be computed for each AY and then the whole reserve.

2.7.1 Partial payments in Fisher-Lange

A problem in F-L methodology can be the presence of partial payments, look at this example:

- 2 claims reported
- 1 paid fully (100€) and 1 paid partially (50€)

The incremental payment will be 150€ but the number of claim paid (n) will be 1, since the partial payments and not considered i the number. For this reason the average cost can be biased (since it should be 75€ and not 150€). The number will be considered when the partial payment will be closed (so when also the other part of the claim will be paid).
Crucial point is: higher is the stability of partial payment and higher will be the quality of estimation. If we are dealing with a LoB with a huge amount of partial payments, we cannot apply the F-L (unless they are very stable).

2.8 Frequency-Severity Methods

Frequency severity methods, takes into account separately frequency and average claim size, in order to estimate the Reserve, as the name suggest. This methodology splits the computation of the incremental payments as the Fisher-Lange methodology does, the difference stands in the coefficient used.
Frequency-Severity used to compute the triangle of Cumulative payments with 2 coefficients: number of paid claims (cumulative), average paid cost (cumulative). While the first coefficient is more intuitive in practise, the second one is

not (given that we never encounter a cumulative average cost). The idea is to have a average cost for cumulative amount (and not for incremental one as we did until now).
One of the classical method uses chain ladder methodology in order to estimate lower triangle for both frequency (number) and severity (cost).
Since the paid C-L is the underlying methodology used, we will end with 2 lambdas computed: one for the frequency and one for the severity.

$$C_{i,j} = n_{i,j}^C \cdot m_{i,j}^C$$

$$n_{i,j}^C = \begin{cases} n_{i,j} & \text{if } j = 0 \\ n_{i,j} + n_{i,j-1}^C & \text{otherwise} \end{cases}$$

$$m_{i,j}^C = \frac{C_{i,j}}{n_{i,j}^C}$$

Once we have computed the upper triangles of number of cumulative numbers and of cumulative average claims size, we apply the chain ladder methodology in order to estimate the lower triangle, the assumptions to be verified for this methodology are the same for chain ladder, The development of the triangles needs to be stable.

$$\widehat{\lambda}_j^n = \frac{\sum_{i=0}^{n-j-1} n_{i,j+1}^C}{\sum_{i=0}^{n-j-1} n_{i,j}^C} \quad \forall j = 0, ..., n-1$$

$$\widehat{\lambda}_j^m = \frac{\sum_{i=0}^{n-j-1} m_{i,j+1}^C}{\sum_{i=0}^{n-j-1} m_{i,j}^C} \quad \forall j = 0, ..., n-1$$

Once the vectors of $\lambda^{\mathbf{m}}$ and $\lambda^{\mathbf{n}}$ are estimated, we can compute the values of the lower triangle.
As usual the reserve will be defined as:

$$R^{FS} = \sum_{i=0}^{n} (\widehat{n}_{i,\infty}^C \cdot \widehat{m}_{i,\infty}^C - C_{i,n-i})$$

In case no tail estimation is needed then the pedix ∞ is equal to n.

Chapter 3

Market Consistent Valuation of Technical Liabilities

3.1 Asset and Liability

Solvency II Directive defines a market consistent valuation of assets and liabilities:

- Assets should be valued at the amount for which they could be exchanged between knowledgeable willing parties in an arm's length transaction;
- Liabilities should be valued at the amount for which they could be transferred, or settled, between knowledgeable willing parties in an arm's length transaction.

While it can be easily found a market value for assets, it cannot for liabilities since it does not exist a market where liabilities are traded. Solvency II tries to define the concept of market consistent evaluation. The directive states that the value of the technical provision should correspond to the amount that I have to pay to another company in order to transfer all the liabilities and this amount is the current exit value of the liability.
Liabilities can be:

- **Hedgeable**: I have a financial instrument in the market with the same characteristics of the liability, so it is possible to use the asset market value to evaluate the liability (Unit-linked products of the life business are a common example). For the moment, there are no headgeable liability for the non-life business
- **Non-Hedgeable**: Best estimate+Risk margin. Company A evaluates the price to transfer its liability to B (company A pays B to give its liability

side to another company). B will ask to A not only the expected value of future cash-out (Best Estimate) but also a remuneration (a margin) for the risk (Risk margin).

3.2 Best Estimate

Best Estimate is calculated on the basis of current and credible information and realistic assumptions, taking account of financial guarantees and options in insurance or reinsurance contracts. The cash-flow projection used in the calculation of the best estimate shall take account of all the cash in- and out-flows. Best estimate must be evaluated for single claim or contract. It is allowed to reduce undue burden on the undertaking: the policies could be grouped and the projection of future cash-flows based on suitable model points could be permitted (taking into account Homogeneous Risk Groups). Best Estimate (BE) must be also evaluated net of reinsurance $BE^{net} = Be^{gross} - recoveries$. Recoveries from reinsurance contracts shall take account of expected losses due to default of the counterparty.

3.2.1 BE - Premium Reserve

The Best Estimate is given by the difference between cash-out and cash-in. In the Life business we do this split also for the local GAAP (since the Mathematical Reserve is the difference between the amount the company expects to pay and the amount received by the policyholder). In the Non-Life business the starting point is the Premium Provision: this reserve serves to cover claim not yet occurred from existing contracts (so claims that could occur in the future).

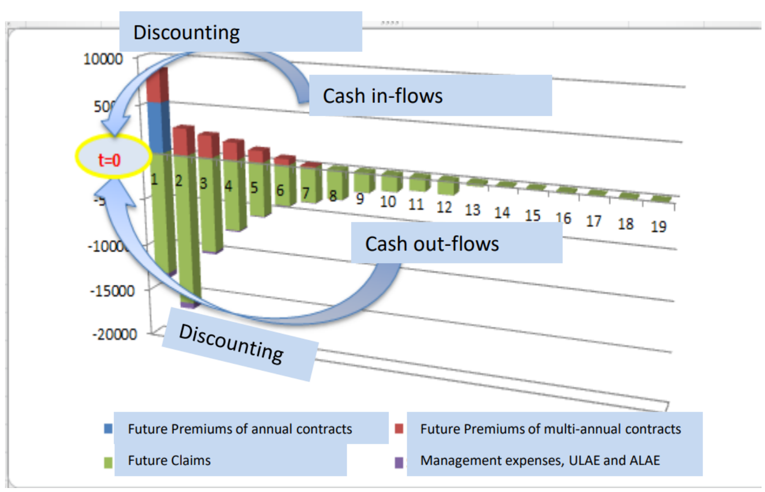

Figure 3.1: ft

The blue bar considers the portion of premium received for annual contracts, the red bars refer to premiums of multi-annual contracts. The green bars are considering future claims and they will stop when the company pays the last claim of the current year (the company needs to evaluate not only the amount to pay but also the time period of that amount given that we need to apply the discounting effect). Larger is the duration of the contract and larger will be the amount of the green bars.
The purple bars consider the expenses (settlement, management and acquisition) that are mostly paid in the first years. Premium Reserve could be lower than the unearned premium reserves (this situation never occurs in the local GAAP).
The formula to follow in order to compute the BE of Premium Provision is:

- The first product is the combined ratio (CR) times the part of the unearned premium reserve (UPR) of the local GAAP (VM): higher is the combined ratio and higher will be the part of the UPR w will put aside;
- (CR-1) times PVFP: the present value of future premiums is important when we have multi-annual contracts. If the part in the brackets is lower than 1 means that in future we will have a profit so we can tke out a part of those future premiums from the value of the BE, the vice-versa if the value in the bracket is higher than 1
- AER times PVFP: AER is a ratio considering only the acquisition expenses (acquisition expenses/premiums), so we are adding to the BE the AC we will have in future for existing contracts.

3.2.2 BE - Claim Reserve

BE of claims reserves is derived as the discounted expected value of payments and expenses (so cash-out) for claims already incurred at the valuation date (including IBNR). It should be evaluated net of recoverables.
Difference with respect to local GAAP

- discounting:
- possible presence of cash-in:

This situation can lead to some judgements to be done from the actuaries: should they consider the claim reserve net or gross of recoverables? In case the recoverables are present into the triangle (gross of rec.) there can be some amount in some cell that does not permit to use some models (for example the Log-Normal distribution since its domain is positive); in case there are no recoverables, they need to be estimated in another triangle (this is the most preferred option by actuaries).

3.3 Discounting Rates

For a given currency and valuation date, each insurance and reinsurance undertaking should use the same relevant basic risk-free interest rate term structures. After the publication of Delegated Acts, EIOPA is doing some refinements (last updates September 2021) to the calculation of interest rate and volatility adjustment.
3 important aspects to take into account when dealing with discounting:

- we need a curve: so a different rate for different duration
- the risk-free rate term structure is different for every currency
- it is risk-free but in the market there are few financial instrument with 0 risk

Risk-free rate term structure is computed considering swap rates (and not government bond rates to avoid concentration risk and liquidity risk: if only one state is used for the computation of risk-free rate then all the insurance system would be affected by the financial condition of that state), govies' rates are considered only if swap rates are not available. Then the CRA (credit risk adjustment) is required, since the swap rates are not risk-free.
The risk-free rate term structure is computed in this way up to the Last liquid point (LLP) set equal to 20 years for Euro currency (50 years for the GBP) and then it is extrapolated from the Ultimate forward rate (UFR) that has a given point of convergence, reached after 60 years.

3.3.1 Volatility Adjustment

The VA is part of the LTGA (long term guaranteed assessment) package in order to avoid pro-cyclical situation in the market: if this measure was not present, insurance companies would sell their assets under some circumstances (i.e. crisis periods) affecting even more the condition of financial market.
VA is an upward shift of the risk-free rate curve (increasing the value of the risk-free rate leads to a major discounting, decreasing the BE value) up to the LLP, after it converges to the UFR.

$$\begin{aligned} VA_{total} = & \underbrace{0.65 \cdot S^{RC}_{currency}}_{\text{Risk Corrected Currency Specific Spread}} + \\ & \underbrace{+\, 0.65 \cdot \overbrace{1_{\left(S^{RC}_{country} > 1\%\right)}}^{\text{Indicator Function}} \cdot \max\left(S^{RC}_{country} - 2 \cdot S^{RC}_{currency}; 0\right)}_{\text{Risk Corrected Country specific Spread}} \end{aligned}$$

Where the S denotes a spread difference and it is computed both for the currency and country parts. While currency spread is always applied, country one is not (country VA kicks in only when $S^{RC}_{country}$ is higher than 85 basis points: this threshold has been changed by the European commission on April 2019).

- $S^{RC}_{currency}$ is computed as the difference between the value of a reference portfolio and the risk-free rate, then a further correction is applied for the usage of swap rates (since they have a risk component);
- $S^{RC}_{country}$ has 2 trigger points: first it has to be greater than 85 bps, second it has to be higher than twice the risk corrected currency spread. If it does not respect these bullets its value is 0;
- VA_{total} is the 0.65 times the sum of these 2 components.

3.3.2 Matching Adjustment

The Matching Adjustment:

- is made in respect of a predictable portfolios of liabilities whose assets are characterized by fixed cash flows and are held to maturity
- is applied as a parallel shift overall basic risk-free curve to decrease the spread of the assets

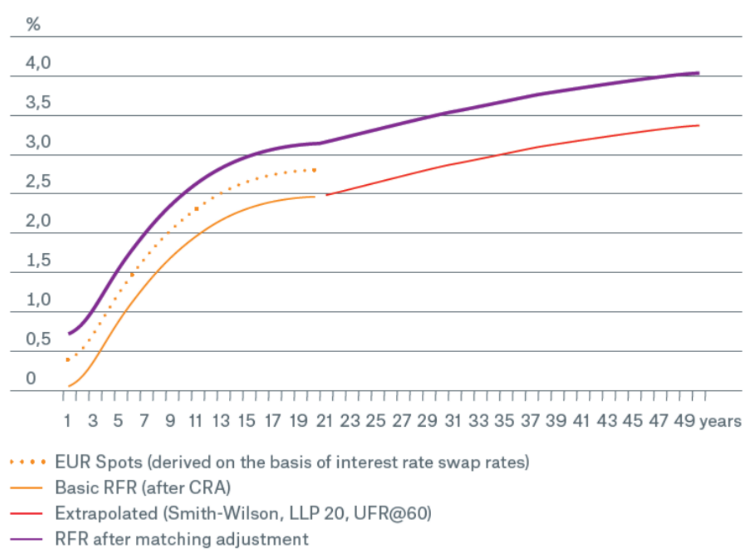

Figure 3.2: Matching Adjustment

- is computed as difference among market rate of asset, risk-free rate and fundamental spread (that is equal to the risk correction for the VA).

3.3.3 Duration

As for a financial asset, it is possible to compute the duration or the convexity of claims reserve. The duration is important to see what is the time in which the payments for existing claims vanishes. MTPL duration is close to 3 (2.85) even if this LoB has long tail. Lower is the duration and less is the insurance company affected by changes of financial conditions (inflation, interest rate and so on).

3.4 Risk Margin

The risk margin is obtained by:

$$RM = \sum_{t=0}^{T} CoC_{rate} \cdot v(0, t+1) \cdot SCR_t$$

- projecting the SCR until the complete run-off of the overall liabilities (diversification assumed),
- quantifying the cost of capital of each year
- discounting using the basic risk-free rates (without corrections for VA or MA).

The SCR consider only risks without a premium risk already computed (market risk is not considered since in the market a remuneration for the risk is already computed).
So it considers: underwriting risk (only for existing business), default risk (with respect to insurance and reinsurance undertakings), operational risk and material risk (only for life business, related to annuities). SCR projection considers also the diversification effect ($SCR_{(MTPL,GTPL)} < SCR_{(MTPL)} + SCR_{(GTPL)}$), but in the market the insurer sells all its Risk Margin, computed for all the Lobs where diversification matters (the difference stands in Life business where diversification is not considered).
Then the discounting of SCR is made for $t + 1$ years, since SCR_0 is what the company need to have at disposal for all the year 1, for this reason it would have no sense to discount it between 0 and t.
The cost of capital (CoC) expresses the the cost of managing liabilities until their run-off. It is fixed at 6 per cent by the regulation since it would change company by company (so to have a comparable value in the market).

3.4.1 Risk Margin proxy

The following simplified methods can be used to derive Risk Margin:

- Approximate the SCR for sub-modules:
- Approximate the whole SCR for each future year, e.g. by using a proportional approach;
- Estimate all future SCRs "at once", e.g. by using an approximation based on the duration approach (used by life insurer);

- Approximate the risk margin by calculating it as a percentage of the best estimate (strong assumption, used by small entities)

QIS5 report shows that: 40 per cent of Non-Life insurers used a proportional approach; 20 per cent of Non-Life insurers used a percentage of best estimate.

Proportional Rule

This simplification is based on the assumption that the future SCRs are proportional to the best estimate of technical provisions for the relevant year. The proportionality factor is computed net of reinsurance (BE considering closed portfolio).

$$SCR_t = SCR_0 \cdot \frac{BE_t^{net}}{BE_0^{net}}$$

The ratio between the Best estimates is computed considering a closed portfolio (since the SCR also is computed in this way). This means that for BE_1^{net} we won't consider claims arising from new contracts and for this reason the BE will reduce with time pass by.
Main drawback is that we consider the evolution of SCR_t exactly equal to the one of BE (if in next year we pay the 50 per cent of claims also the SCR moves in the same direction.

Chapter 4

Stochastic Models for Claims Reserve

4.1 Introduction

Stochastic models serve for associate the uncertainty to claim reserving (focusing especially on variability or on skewness), following a total run-off approach. They can be used also to computed Capital requirement (so in a one-year approach).

4.2 Mack Model

Mack Model Extend the Chain-Ladder idea in a stochastic Framework, it is considered to be the breakthrough in stochastic claims reserving modelling.
The main objective of the model is to take into account the volatility of the Reserve, by calculating the prediction error of the estimated Reserve. In the model we have two different sources of volatility:

- Process Variance: is the variance originated from the Stochastic movements of the process;
- Estimation Variance: reflects the variability/uncertainty induced by the use of a specific estimator.

Assumptions:
There exist parameter λ_j and σ_j such that, for each $i = 0, ..., n$, and $j = 0, ..., n-1$ the following relations hold:

- $\mathbb{E}(C_{i,j+1}|C_{i,j}) = C_{i,j} \cdot \lambda_j$, which implies that $\lambda_j = \mathbb{E}\left(\frac{C_{i,j+1}}{C_{i,j}}|C_{i,1}, ..., C_{i,j}\right)$, that is the classical chain ladder assumption;
- $Var(C_{i,j+1}|C_{i,j}) = C_{i,j} \cdot \sigma_j^2$

- Cumulative payments of different AY (rows) are independent

$$(C_{i,0}, C_{i,1}, ..., C_{i,n}) \perp\!\!\!\perp (C_{j,0}, C_{j,1}, ..., C_{j,n}) \quad \forall i \neq j$$

As a consequence, Cumulative Payments form a Markov Chain for each Accident Year (row i).
The only source of information for claims prediction is the upper triangle data, namely filtration $\mathbf{D^{(n)}} = \{C_{i,j} : i + j \leq n\}$.
Chain Ladder factors are estimated as:

$$\widehat{\lambda}_j = \frac{\sum_{i=0}^{n-j-1} w_{i,j} \cdot f_{i,j}}{\sum_{i=0}^{n-j-1} w_{i,j}} \quad for \quad j = 0, 1, ..., n-1$$

where $f_{i,j} = \frac{C_{i,j+1}}{C_{i,j}}$, and $w_{i,j}$ is a weight, in Mack model, the estimation error derived use as weights $C_{i,j}$ (Classical Paid Chain Ladder).
For the Variance:

$$\widehat{\sigma}_j^2 = \frac{1}{n-j-1} \cdot \sum_{i=0}^{n-j-1} w_{i,j} \cdot (f_{i,j} - \widehat{\lambda}_j)^2 \quad for \quad j = 0, 1, ..., n-2$$

In order to compute the term $\widehat{\sigma}_{n-1}^2$, Mack proposed the following solution:

$$\widehat{\sigma}_{n-1}^2 = \min\left(\frac{\widehat{\sigma}_{n-2}^4}{\widehat{\sigma}_{n-3}^2}, \widehat{\sigma}_{n-2}^2, \widehat{\sigma}_{n-3}^2\right)$$

Which tries to approximate a decreasing exponential pattern. Other methods proposed are based on a smoothing procedure (parametric curve).
PROPERTIES OF THE ESTIMATORS:
Under the model assumptions, it is possible to prove that:

- $\mathbb{E}(\widehat{\lambda}_j | D^{(n)}) = \lambda_j \quad \forall\ j$ Estimator is **Conditionally Unbiased**. Implying $\mathbb{E}(\hat{C}_{i,n} | D^{(n)}) = C_{i,n} \quad \forall\ j$

- $\mathbb{E}(\widehat{\lambda}_j) = \lambda_j \quad \forall\ j$ Estimator is **Unconditionally Unbiased**.

- $\mathbb{E}(\widehat{\lambda}_j \cdot \widehat{\lambda}_k) = \lambda_j \cdot \lambda_k \quad \forall\ j \neq k$ Estimators are **Uncorrelated** (warning: uncorrelation does not imply independence).

- In case $w_{i,j} = C_{i,j}$ estimators of λ_j are **Optimal**, meaning having minimal conditional variance, among all other possible estimators (of the same class). $Var(\widehat{\lambda}_j | D^{(n)}) = \min\limits_{\alpha_{i,j} \in \mathbb{R}} \sum\limits_{i=0}^{n-j-1} \alpha_{i,j} \cdot f_{i,j}$ is minimized by $\alpha_{i,j} = \dfrac{C_{i,j}}{\sum\limits_{i=0}^{n-j-1} C_{i,j}}$

- estimators of σ_j^2 are **Conditionally Unbiased**

- estimators of σ_j^2 are **Unconditionally Unbiased**

4.2.1 Prediction Variance

For a generic random variable y and its predicted value $\widehat{y}$ we have:

$$msep(\widehat{y}) = \mathbb{E}\left((y-\widehat{y})^2\right) = \mathbb{E}\left\{[(y-\mathbb{E}(y)) - (\widehat{y}-\mathbb{E}(y)]^2\right\} =$$

$$= \mathbb{E}\left((y-\mathbb{E}(y))^2\right) + \mathbb{E}\left((\widehat{y}-\mathbb{E}(y))^2\right) - 2\cdot\mathbb{E}\left((y-\mathbb{E}(y))\cdot(\widehat{y}-\mathbb{E}(y))\right)$$

If we introduce $\mathbb{E}(y)=\mathbb{E}(\widehat{y})$, and we neglect the double product, we can rewrite it as an **approximation**:

$$\underbrace{\mathbb{E}\left((y-\widehat{y})^2\right)}_{Prediction\ \ Variance} \approx \underbrace{\mathbb{E}\left((y-\mathbb{E}(y))^2\right)}_{Process\ \ Variance} + \underbrace{\mathbb{E}\left((\widehat{y}-\mathbb{E}(\widehat{y}))^2\right)}_{Estimation\ \ Variance}$$

which is just the sum of the variances, **Process Variance** and **Estimation Variance**.

In our context, the Mean square Error Prediction (MSEP) for an accident year's Reserve ($\hat{R}_i$), given the filtration $D^{(n)}$ is composed by:

$$msep(\hat{R}_i|D^{(n)}) = msep(\hat{C}_{i,n}|D^{(n)}) = \mathbb{E}\left[\left(C_{i,n}-\hat{C}_{i,n}\right)^2|D^{(n)}\right] =$$

$$= \mathbb{E}\left[\left(C_{i,n}-\mathbb{E}(C_{i,n}|D^{(n)})\right)^2|D^{(n)}\right] + \mathbb{E}\left[\left(\hat{C}_{i,n}-\mathbb{E}(C_{i,n}|D^{(n)})\right)^2|D^{(n)}\right]$$

$$= \underbrace{Var(C_{i,n}|D^{(n)})}_{\text{Process Variance}} + \underbrace{\left(\hat{C}_{i,n}-\mathbb{E}(C_{i,n}|D^{(n)})\right)^2}_{\text{Estimation Variance}}$$

Estimation Variance does not include the expected value, because both $\hat{C}_{i,n}$ and $\mathbb{E}(\hat{C}_{i,n}|D^{(n)})$ are **assumed to be deterministic conditional on data**.

4.2.2 Process Variance

Process variance is computed as:

$$Var(C_{i,n}|D^{(n)}) = Var(C_{i,n}|C_{i,n-i}) = \boxed{C_{i,n-i}\cdot\sum_{m=n-i}^{n-1}\left(\sigma_m^2\cdot\prod_{h=n-i}^{m-1}\lambda_h\cdot\prod_{k=m+1}^{n-1}\lambda_k^2\right)}$$

The Proof is given in 7.1

To obtain an estimation of the process Variance, we now have to include the estimators for the unknown quantities σ_m^2, and λ_j.

$$\widehat{Var}(C_{i,n}|C_{i,n-i}) = C_{i,n-i}\cdot\sum_{m=n-i}^{n-1}\left(\prod_{h=n-i}^{m-1}\widehat{\lambda}_h\cdot\widehat{\sigma}_m^2\cdot\prod_{k=m+1}^{n-1}\widehat{\lambda_k^2}\right)$$

$$= \sum_{m=n-i}^{n-1}\hat{C}_{i,m}\cdot\frac{\sigma_m^2}{\widehat{\lambda}_m^2}\cdot\prod_{k=m}^{n-1}\widehat{\lambda_k^2} = \boxed{\hat{C}_{i,n}^2\cdot\sum_{m=n-i}^{n-1}\frac{\widehat{\sigma}_m^2}{\widehat{\lambda}_m^2\cdot\widehat{C}_{i,m}}}$$

It is worth to notice that for $\hat{C}_{i,m}$, when $i+m \leq n$ then $\hat{C}_{i,m} = C_{i,m} \in \quad D^{(n)}$. Another way to compute the variance is in a recursive way:

$$\widehat{Var}(C_{i,j}|D^{(n)}) = \widehat{\sigma}^2_{j-1} \cdot \hat{C}_{i,j-1} + \widehat{\lambda}^2_{j-1} \cdot \widehat{Var}(C_{i,j-1}|D^{(n)})$$

Given the independence between AY (rows) of the Cumulative payments:

$$\widehat{Var}(R|D^{(n)}) = \sum_{i=1}^{n} \widehat{Var}(R_i|D^{(n)}) = \sum_{i=1}^{n} \hat{C}^2_{i,n} \cdot \sum_{m=n-i}^{n-1} \left(\frac{\widehat{\sigma}^2_m}{\widehat{\lambda}^2_m \cdot \hat{C}_{i,m}} \right)$$

In order to compute the Coefficient of Variability (CoV) of the claims reserve (considering only process variance):

$$CoV(R|D^{(n)}) = \frac{\sqrt{\sum_{i=1}^{n} \hat{C}^2_{i,n} \cdot \sum_{m=n-i}^{n-1} \left(\frac{\widehat{\sigma}^2_m}{\widehat{\lambda}^2_m \cdot \hat{C}_{i,m}} \right)}}{\sum_{i=1}^{n} \hat{C}_{i,n} - C_{i,n-i}}$$

4.2.3 Estimation Variance

We focus now on the second term of the *msep*, the Estimation Variance:

$$\left(\hat{C}_{i,n} - \mathbb{E}(C_{i,n}|D^{(n)})\right)^2 = \left(C_{i,n-i} \cdot \prod_{t=n-i}^{n-1} \widehat{\lambda}_t - \prod_{t=n-i}^{n-1} \lambda_t \right)^2$$

The term involving the real values of λ, which are not known, needs to be estimated in order to estimate how far from the real values are. Several approaches have been proposed, and the one illustrated below, is the one proposed by Merz and Wutrich.
We can rewrite the formula as a telescoping sum:

$$C^2_{i,n-i} \cdot \left(\prod_{t=n-i}^{n-1} \widehat{\lambda}_t - \prod_{k=n-i}^{n-1} \lambda_k \right)^2 = C^2_{i,n-i} \cdot \left(\sum_{k=n-i}^{n-1} S_k \right)$$

Where the term s_k is:

$$s_k = \prod_{t=n-i}^{k-1} \widehat{\lambda}_t \cdot (\lambda_k - \widehat{\lambda}_k) \cdot \prod_{t=k+1}^{n-1} \lambda_t$$

$$\left(\hat{C}_{i,n} - \mathbb{E}(C_{i,n}|D^{(n)}\right)^2 = \boxed{\hat{C}^2_{i,n} \cdot \sum_{k=n-i}^{n-1} \left(\frac{\widehat{\sigma}^2_k}{\widehat{\lambda}^2_k \cdot \sum_{i=0}^{n-k-1} C_{i,k}} \right)}$$

4.2.4 Prediction Error

So for a single AY i, we obtained:

- Process Variance: $\hat{C}_{i,n}^2 \cdot \sum_{m=n-i}^{n-1} \frac{\widehat{\sigma}_m^2}{\widehat{\lambda}_m^2 \cdot \widehat{C}_{i,m}}$

- Estimation Variance: $\hat{C}_{i,n}^2 \cdot \sum_{k=n-i}^{n-1} \left(\frac{\widehat{\sigma}_k^2}{\widehat{\lambda}_k^2 \cdot \sum_{i=0}^{n-k-1} C_{i,k}} \right)$

So for msep we have:

$$msep(\hat{R}_i|D^{(n)}) = \hat{C}_{i,n}^2 \cdot \left(\sum_{m=n-i}^{n-1} \frac{\widehat{\sigma}_m^2}{\widehat{\lambda}_m^2 \cdot \widehat{C}_{i,m}} + \sum_{k=n-i}^{n-1} \left(\frac{\widehat{\sigma}_k^2}{\widehat{\lambda}_k^2 \cdot \sum_{i=0}^{n-k-1} C_{i,k}} \right) \right) =$$

$$= \hat{C}_{i,n}^2 \cdot \sum_{j=n-i}^{n-1} \frac{1}{\widehat{\lambda}_j^2} \cdot \left(\overbrace{\frac{\widehat{\sigma}_j^2}{\hat{C}_{i,j}}}^{\widehat{Var}(f_{i,j})} + \overbrace{\frac{\widehat{\sigma}_j^2}{\sum_{i=0}^{n-j-1} C_{i,j}}}^{\widehat{Var}(\widehat{\lambda}_j)} \right)$$

The first term in the most inner parenthesis is the $\widehat{Var}(f_{i,j})$, while the second term is $\widehat{Var}(\widehat{\lambda}_j)$.
Also the msep can be computed in a iterative way, leading to the possibility of adding a tail factor.

$$msep(\hat{C}_{i,j+1}|D^{(n)}) = \hat{C}_{i,j} \cdot \Big(Var(f_{i,j}) + Var(\widehat{\lambda}_j) \Big) + msep(\hat{C}_{i,j}|D^{(n)})$$

When we need to compute the total msep for claims reserve, we need to take into account also the dependency between the Accident Years:

$$msep(\hat{R}|D^{(n)}) = \sum_{i=1}^{n} msep(\hat{R}_i|D^{(n)}) + 2 \cdot \sum_{i=1}^{n-1} \left[\hat{C}_{i,n} \cdot \left(\sum_{h=i+1}^{n} \hat{C}_{h,n} \right) \cdot \sum_{j=n-i}^{n-1} \left[\frac{\widehat{\sigma}_j^2}{\widehat{\lambda}_j^2 \cdot \sum_{i=0}^{n-j-1} C_{i,j}} \right] \right]$$

Implying that

$$Cov(\hat{C}_{h,n}, \hat{C}_{i,n}) = \left(\frac{\hat{C}_{h,n}}{\hat{C}_{i,n}} \right) \cdot Var(\hat{C}_{i,n}|D^{(n)})$$

for every $h > i$.
We can see how the second factor of the msep is given by a formula similar to that of the Estimation Variance, this is due to the fact that the dependence is given by the same factors used $\widehat{\lambda}_j$ in order to compute $\hat{C}_{i,n}$ for every row, so induces a dependence between rows.

4.3 Over-Dispersed Poisson Model

Over-dispersed Poisson distribution is a Poisson distribution in which the variance is different from the mean.
In Claims reserving the over-dispersed Poisson model assumes that the incremental claims $P_{i,j}$ are distributed as independent over-dispersed Poisson random variables with mean and variance:

- $\mathbb{E}[P_{i,j}] = x_i \cdot y_j$, with $\sum_{j=0}^{n} y_j = 1$, and $y_j > 0$, $\forall j$;
- $Var[P_{i,j}] = \phi \cdot x_i \cdot y_j$.

From the independence of $P_{i,j}$ distributed as Poisson, we can state that also $C_{i,j}$ are Poisson distributed, and in this case we have:

- $\mathbb{E}[C_{i,j}] = x_i$
- $\mathbb{E}\left[\frac{P_{i,j}}{P_{i,h}}\right] = \frac{y_j}{y_h} \quad \forall\, j, h$

In this formula, the mean has a multiplicative structure, in the sense that it is the product of the row effect and column effect. Both x_i and y_j have a specific interpretation, x_i is the **Expected Ultimate Cost**, while y_j is the **proportion of ultimate claim to be imputed** for each development year.
For estimation purposes, it is often better to re-parameterize the model in a linear form of the mean, by using a GLM with a log-link function (also called Logit): $log(m_i, j) = c + \alpha_i + \beta_j$.
Given the original formulation, we can define the likelihood function as:

$$L(x_0, ..., x_n, y_0, ..., y_n | D^{(n)}) = \prod_{i+j \leq n} e^{-x_i \cdot y_j} \cdot \frac{(x_i \cdot y_j)^{P_{i,j}}}{P_{i,j}!}$$

Then passing to the log-likelihood, by applying the logarithm we ovtain:

$$l(x_i, y_j) = ln(L(x_0, ..., x_n, y_0, ..., y_n | D^{(n)}) = \sum_{i=0}^{n} \sum_{j=0}^{n-i} (-x_i \cdot y_j) + P_{i,j} \cdot ln(x_i, y_j) - ln(P_{i,j}!)$$

AAnd as usual, by calculating the $2n+2$ partial derivatives for each x_i and each y_j and equaling them to zero, we get the Maximum Likelihood Estimation

$$\frac{\partial L(x_i, y_j)}{\partial x_i} = \sum_{j=0}^{n-i} (-y_j + y_j \cdot \frac{P_{i,j}}{x_i \cdot y_j}) = 0 \qquad \frac{\partial L(x_i, y_j)}{\partial y_j} = \sum_{i=0}^{n-j} (-x_i + x_i \cdot \frac{P_{i,j}}{x_i \cdot y_j}) = 0$$

We can easily obtain

$$\hat{x}_i \cdot \sum_{j=0}^{n-i} \hat{y}_j = \sum_{j=0}^{n-i} P_{i,j} \qquad \hat{y}_j \cdot \sum_{i=0}^{n-j} \hat{x}_i = \sum_{i=0}^{n-j} P_{i,j}$$

By using this two equations we can show that Chain Ladder Development Factors coincide with the MLE:

$$\hat{C}_{i,n} = C_{i,n-i} + \mathbb{E}\left(\sum_{j=n-i+1}^{n} P_{i,j}\right) \overbrace{=}^{\text{First Equation}} C_{i,n-i} + \hat{x}_i \cdot \sum_{j=n-i+1}^{n} \hat{y}_j =$$

$$= C_{i,n-i} \cdot \left(1 + \frac{\cancel{\hat{x}_i} \sum_{j=n-i+1}^{n} \hat{y}_j}{\cancel{\hat{x}_i} \sum_{j=0}^{n-i} \hat{y}_j}\right) = C_{i,n-i} \cdot \left(\frac{\sum_{j=0}^{n-i} \hat{y}_j + \sum_{j=n-i+1}^{n} \hat{y}_j}{\sum_{j=0}^{n-i} \hat{y}_j}\right) = C_{i,n-i} \cdot \left(\frac{\sum_{j=0}^{n} \hat{y}_j}{\sum_{j=0}^{n-i} \hat{y}_j}\right)$$

$$= C_{i,n-i} \cdot \left(\frac{\sum_{j=0}^{n-i+1} \hat{y}_j}{\sum_{j=0}^{n-i} \hat{y}_j}\right) \cdot \left(\frac{\sum_{j=0}^{n-i+2} \hat{y}_j}{\sum_{j=0}^{n-i+1} \hat{y}_j}\right) \cdot \ldots \cdot \left(\frac{\sum_{j=0}^{n} \hat{y}_j}{\sum_{j=0}^{n-1} \hat{y}_j}\right)$$

then starting from the Chain Ladder Development Factors:

$$\widehat{\lambda}_j = \frac{\sum_{i=0}^{n-j-1} C_{i,j+1}}{\sum_{i=0}^{n-j-1} C_{i,j}} = \frac{\sum_{i=0}^{n-j-1} \sum_{h=0}^{j+1} P_{i,h}}{\sum_{i=0}^{n-j-1} \sum_{h=0}^{j} P_{i,h}} = \frac{\sum_{h=0}^{j+1} \sum_{i=0}^{n-j-1} P_{i,h}}{\sum_{h=0}^{j} \sum_{i=0}^{n-j-1} P_{i,h}} \overbrace{=}^{\text{Second Equation}}$$

$$= \frac{\sum_{h=0}^{j+1} \hat{y}_h \cdot \cancel{\sum_{i=0}^{n-j-1}} \hat{x}_i}{\sum_{h=0}^{j} \hat{y}_h \cdot \cancel{\sum_{i=0}^{n-j-1}} x_i} = \frac{\sum_{h=0}^{j+1} \hat{y}_h}{\sum_{h=0}^{j} \hat{y}_h}$$

So we can see that the terms coming from the Chain Ladder $\widehat{\lambda}_j$ is the generic member of the formula of $\hat{C}_{i,n}$, which is the **Maximum Likelihood Estimator**, given a Poisson distribution with mean equal to $x_i \cdot y_j$, (here the standard deviation is not important).

$$\hat{C}_{i,n}^{ODP} = \frac{C_{i,n-i}}{\sum_{j=0}^{n-i} \hat{y}_j} = \hat{C}_{i,n} = C_{i,n-i} \cdot \prod_{j=n-i}^{n-1} \widehat{\lambda}_j$$

$$\sum_{j=0}^{n-i} \hat{y}_j = \frac{1}{\prod_{i=n-i}^{n-1} \widehat{\lambda}_j} = \frac{1}{\widehat{CDF}_{n-i}}$$

4.4 Bootstrap for Claims Reserve

Bootstrapping techniques are widely used in many fields of statistics. It is the practice of estimating properties of an estimator by measuring the properties

when sampling from an approximating distribution.
One standard choice for an approximating distribution is the empirical distribution of observed data (non-parametric).
In our case we are interested in deriving the distribution of our interest estimators (expected value and Variance, but also other such as skewness or kurtosis) from the deterministic Triangle. With Bootstraping we can derive the **Estimation Variance**.
The main assumption of this technique is the fact that the observed data is the realization of i.i.d. (independent and identically distributed) random variables. Since with our triangle we do not have i.i.d. random variables, we need a way to get them. First 9 points will refer to EV, from 10-12 we refer to PV. This technique is composed by an iterative procedure:

1. Estimate Chain Ladder development Factors $\widehat{\lambda}_j$ as usual :

$$\widehat{\lambda}_j = \frac{\sum_{i=0}^{n-j-1} C_{i,j+1}}{\sum_{i=0}^{n-j-1} C_{i,j}}$$

2. Build a backfitted triangle of observation; starting from the real value of the diagonal $C_{i,n-i}$, and apply recursively the factors λ_j until the first observation $\dot{C}_{i,0}$:

$$\dot{C}_{i,j} \begin{cases} C_{i,n-i} & if \;\; j = n-i \\ \frac{C_{i,n-i}}{\prod_{h=j}^{n-i} \widehat{\lambda}_h} = \frac{\dot{C}_{i,j+1}}{\widehat{\lambda}_j} & if \;\; j < n-i \end{cases}$$

   In this way we will get a pseudo-cumulative triangle (only upper part). A particular result is given for the penultimate diagonal where Pseudo-cumulative values will be equal to cumulative ones when the lambda is equal to the individual link ratio. In the previous diagonal this situation does not hold since different lambdas are present, so the pseudo-cumulative values would be different from the $C_{i,n-i}$ even if the individual link ratio is equal to lambda.

3. From pseudo-cumulative values we build the triangle of pseudo-incremental Paid claims:

$$\dot{P}_{i,j} = \begin{cases} \dot{C}_{i,0} & if \;\; j = 0 \\ \dot{C}_{i,j} - \dot{C}_{i,j-1} & if \;\; 0 < j \leq n-i \end{cases}$$

4. Compute the Pearson Residuals

$$r_{i,j} = \frac{P_{i,j} - \dot{P}_{i,j}}{\sqrt{\dot{P}_{i,j}}}$$

We compute them since we need i.i.d. random variables to apply the methodology. Residuals are commonly used in order to exploit i.i.d. properties.

5. Adjust the Residuals:

$$r_{i,j}^{adj} = r_{i,j} \cdot \sqrt{\frac{\frac{N(N+1)}{2}}{\frac{N(N+1)}{2} - (2N-1)}}$$

Where the term $\frac{N(N+1)}{2}$ is the number of elements in the upper triangle, and $2N-1$ is the number of parameters ($2N-1$ means $2(n+1)-1$ so it is the number of parameters to be estimated. If our triangle is 3x3 (so $N=4$), we have to estimate 8 parameters, but since the underlying distribution is the ODP and one assumption is $\sum_{j=0}^{n} y_j = 1$, we can estimate 7 parameters and the 8^{th} is obtained as $1-$(sum of the seven parameters)), and $N = n+1$ is the number of rows (or columns if the triangle is squared). We adjust residuals since the sample is not so huge and we know that the i.i.d. r.v. properties hold (at least asymptotically) if the sample is large enough.

6. Reiterate Steps from 7 to 9 for $h = 1, ..., H$:

7. Resample the **adjusted residuals** with replacement, creating a new past triangle of adjusted residuals $r_{i,j}^{adj,h}$

8. Build the triangle of past incremental values starting from the residuals with the following equation:

$$P_{i,j}^{h} = \dot{P}_{i,j} + r_{i,j}^{adj,h} \cdot \sqrt{\dot{P}_{i,j}}$$

9. build the triangle of cumulative data $C_{i,j}^{h}$, apply the classical chain ladder in order to compute R_i^h, and R^h, and save this two amounts.

Now we have H different values of R^h and R_i^h, so we can see these values as an empirical distribution of the Reserve, used for compute the desired quantities (such as the variance, but also skewness, kurtosis and other useful indicators). The expectation of the values should be similar to the standard deterministic value of the Reserve, otherwise we have an high Bias in our model.

4.4.1 Including Process Variance in Bootstrap

Different ways are possible in order to include the process variance in this technique

Closed Formula:
the first one is by using a closed formula described by Verrall and England:

$$msep(\hat{R}|D^{(n)}) = \phi \cdot \hat{R} + Var_{bs}(\hat{R})$$

in which $Var_{bs}(\hat{R})$ is the variance using the bootstraping method, and ϕ is a scaling parameter used for the process variance:

$$\widehat{\phi} = \frac{\sum_{i=0}^{n} \sum_{j=0}^{n-i} r_{i,j}^2}{\frac{N(N+1)}{2} - (2N-1)}$$

This result can be derived by showing that:

$$\sum \left(\frac{P_{i,j} - \dot{P}_{i,j}}{\sqrt{\dot{P}_{i,j}}} \right)^2 \sim \chi^2_{df}$$

with degrees of freedom equal to $df = \frac{N(N+1)}{2} - (2N-1)$
Simulation Approach: Another way to include the process variance in this methodology is by means of simulation, by extending the original Bootstrapping technique, in order to provide a way of simulating a **complete predictive distribution**.
This formula is not widely used since we want a methodology to derive the distribution for Reserve but doing in this way we do not obtain any distribution, we simply get a value for the *msep*.
Simulation Approach In order to include variability of the process into the bootstrap technique we can add some steps in order to compute a simulation of the values from a given distribution:
Apply steps 1 to 9 of bootstrap, in order to obtain H lower triangles of pseudo-cumulative paid amounts: $\hat{C}^h_{i,j}$, with $(i + j > n)$.

10. Obtain the corresponding H future triangles of incremental payments $\hat{P}^h_{i,j}$ in each cell $(i+j > n)$ by differencing. These values are used as the mean of the **Poisson** distribution for simulating.

11. For each cell i, j of the lower triangles $(i+j > n)$ in the future triangle, for each simulation h, we extract a payment $\ddot{P}^h_{i,j}$ from a Poisson with mean equal to $\hat{P}^h_{i,j}$, and variance $\phi \cdot \hat{P}^h_{i,j}$, given the fact that for a poisson distribution mean and variance are equal to the parameter of the distribution, in order to obtain values with mean equal to $\hat{P}^h_{i,j}$ and variance $\phi \cdot \hat{P}^h_{i,j}$, we extract $\ddot{y}^h_{i,j} \sim Pois(\frac{\hat{P}^h_{i,j}}{\phi})$, and then we multiply $\phi \cdot \ddot{y}^h_{i,j}$, in order to get $\ddot{P}^h_{i,j}$, so that Expected values and variance will be equal to the desired ones. In practice we simulate $\ddot{y}^h_{i,j}$ from a $Pois(\frac{\hat{P}^h_{i,j}}{\phi})$, and then $\ddot{P}^h_{i,j} = \phi \cdot \ddot{y}^h_{i,j}$

12. Sum the values in the lower triangle to get the value of the reserve, both for Accident Year R_i^h, and total R^h.

In step 11, if $\hat{P}_{i,j}^h$ is negative, then we cannot simulate directly using a Poisson, which has a positive Support, indeed we need to adjust our formula: $\ddot{y}_{i,j}^h \sim Pois\left(\left|\frac{\hat{P}_{i,j}^h}{\phi}\right|\right)$, and then we put $\ddot{P}_{i,j}^h = \phi \cdot \hat{y}_{i,j}^h - 2\max(-\hat{P}_{i,j}^h, 0)$, this is not a real solution, but it can be used if not many cells are negative, and if the amount is not very big (in absolute term).

4.5 Collective Risk Model

Is a frequency Severity type of stochastic model, used in premium risk context in order to describe the aggregate claim amount. Under Classical assumptions, the amount of claims outstanding is defined as:

$$X = \sum_{h=1}^{K} Z_h$$

where:

- X is the r.v. that describes the total amount of claims already incurred that will be paid in the future (Claims Reserve);
- K is the r.v. that describes the number of claims;
- Z_h is the r.v. that describes the average cost.

As usual we assume that K is independent from Z_h, for every h, and Z_h are i.i.d.
The main **Drawback** of this formulation is that we don't have any distinction for amounts by Accident nor Development Year. So a generalization has been proposed in order to take into account each amount in the lower triangle to be described by a Collective Risk Model:

$$X_{i,j} = \sum_{h=1}^{K_{i,j}} Z_{i,j,h}$$

where:

- $X_{i,j}$ is the amount in each cell of the lower triangle;
- $K_{i,j}$ is the r.v. claims count of the i-th AY and effectively paid in the future j-th development year;
- $Z_{i,j,h}$ is the r.v. paid claims amount of the h-th single claim of i-th AY paid in the j-th DY.

Also in this case the main assumptions are independence between $K_{i,j}$ and $Z_{i,j,h}$, for every h, $Z_{i,j,h}$ i.i.d.
Once the assumptions have been fulfilled, it is possible to obtain the exact moments of cumulants for each cell of the triangle. One of the main difficulties of the model comes from the calibration of the parameters, given the fact that for each cell we have only one realization.
The random Variable $K_{i,j}$ is assumed to be distributed as a Poisson,

$$K_{i,j} \sim Poisson\left(\hat{n}_{i,j} \cdot q\right)$$

where $\hat{n}_{i,j}$ is the expected number of claims in cell i,j coming from a deterministic method, and q is a random variable used to catch the possible fluctuations around the expected value computed using a deterministic method; $\mathbb{E}(q) = 1$, $q > 0$, and $Var(q) = \sigma_q^2$. The disturbance factor q can also be used singularly for each cell $q_{i,j}$, but in this case we will not catch the dependence between the different parts of the triangle. Also q_j is possible to be used, which will catch the dependence between different rows.
The Variable $Z_{i,j,h}$ is assumed to be distributed with a mean equal to $\hat{m}_{i,j}$ which is the average cost estimate by a deterministic method, and coefficient of variability $C_{Z_{i,j,h}} = C_{Z_j} \cdot r_j \quad \forall i$. Also in this case we can introduce a parameter catching the uncertainty of the estimation by using r_j, that changes the coefficient of variability of the variable $Z_{i,j}$. r_j is assumed to be positive, and with $\mathbb{E}(r_j) = 1$, also we assume that r_j is independent from r_k for $j \neq k$, and q independent from r_j. r_j produces a positive association between rows. The estimation of $C_{Z_{i,j}}$ is done by using the data from the single claims, in order to obtain the triangle of $\hat{C}_{Z_{i,j}}$. $\hat{C}_{x_j}$ will be the weighted average of each column of $\hat{C}_{Z_{i,j}}$.
By having the data of the single claim we could in theory fit for each cell a different distribution, but generally this is a problem when we have to choose the distribution of the lower triangle $Z_{i,j,h}$, so it is usual in practice to use the same distribution for every cell, or to use at least the same distribution for the same column.
Under previous assumptions and assuming that each r_j is distributed as a Gamma(h,h), it is possible to prove the following relations:

$$\begin{aligned}
\mathbb{E}(X_{i,j}) &= \hat{n}_{i,j} \cdot \hat{m}_{i,j} = \hat{X}_{i,j} \\
Var(X_{i,j}) &= \hat{n}_{i,j} \cdot \hat{\alpha}_{2,Z_{i,j}} + \hat{n}_{i,j}^2 \cdot \hat{m}_{i,j}^2 \cdot \sigma_q^2 + \hat{n}_{i,j} \cdot \hat{\sigma}_{Z_{i,j}}^2 \cdot \sigma_{r_j}^2 \\
CV(X_{i,j}) &= \sqrt{\frac{1+\hat{C}_{Z_{i,j}}^2}{n_{i,j}} + \sigma_q^2 + \frac{\hat{C}_{Z_{i,j}}^2 \cdot \sigma_{r_j}^2}{n_{i,j}}} = \sqrt{\frac{1+\hat{C}_{Z_{i,j}}^2 \cdot \alpha_{2,r_j}}{n_{i,j}} + \sigma_q^2}
\end{aligned}$$

where:

$$\hat{\sigma}_{Z_{i,j}}^2 = \hat{m}_{i,j}^2 \cdot \hat{C}_{Z_{i,j}}^2 \quad and \quad \hat{\alpha}_{2,Z_{i,j}} = \hat{\sigma}_{Z_{i,j}}^2 + \hat{m}_{i,j}^2$$

We can see that $\lim_{n_{i,j} \to +\infty} CV(X_{i,j}) = \sqrt{\sigma_q^2} = \sigma_q$, but in this case given the fact that $n_{i,j}$ here is the number of claims in one cell, it is difficult to observe values

very high, and for this reason the diversification effect is always smaller then the one in Pricing or Premium Reserve.
Assuming that $i = 0, 1, ..., n$ and $j = 0, 1, ..., n^+$, the estimate of claims reserve is equal to:

$$\hat{R} = \mathbb{E}\left(\sum_{i=0}^{n}\sum_{j=n-i+1}^{n^+} X_{i,j}\right) = \sum_{i=0}^{n}\sum_{j=n-i+1}^{n^+} \hat{n}_{i,j}\cdot\hat{m}_{i,j}$$

The Variance of the Claims Reserve, neglecting the r.v. r_j, is given by:

$$\sigma^2(\hat{R}) = \sum_{i=0}^{n}\sum_{j=n-i+1}^{n^+} \hat{n}_{i,j}\cdot\hat{\alpha}_{2,Z_{i,j}} + \overbrace{\left[\sum_{i=0}^{n}\sum_{j=n-i+1}^{n^+} \hat{n}_{i,j}\cdot\hat{m}_{i,j}\right]^2}^{(\mathbb{E}(R))^2}\cdot\sigma_q^2$$

Where the first term gives the sum of the variances of different cells, given independence, while the second term catches the dependence between the claims count of different cells, given by q.
For the general idea between the variance see proof 7.3.
In order to include also the term r_j, we should have another term involving the additional variance, and another one involving the covariance induced by the random variable r_j.
Given that closed formula solution are very difficult to get, so for further moments simulation is usually preferred.
For a single AY i we get:

$$\sigma^2(\hat{R}_i) = \sum_{j=n-i+1}^{n^+} \hat{n}_{i,j}\cdot\hat{\alpha}_{2,Z_{i,j}} + \mathbb{E}(\hat{R}_i)^2\cdot\sigma_q^2$$

And the covariance between two generic cells is given by:

$$Cov(X_{i,j}; X_{h,k}) = \hat{n}_{i,j}\cdot\hat{m}_{i,j}\cdot\hat{n}_{h,k}\cdot\hat{m}_{h,k}\cdot\sigma_q^2 = \hat{P}_{i,j}\cdot\hat{P}_{h,k}\cdot\sigma_q^2$$

$$\rho = \frac{Cov(X_{i,j}, C_{h,k})}{\sigma(X_{i,j})\cdot\sigma(X_{h,k})} = \frac{\hat{P}_{i,j}\cdot\hat{P}_{h,k}\cdot\sigma_q^2}{\sigma(X_{i,j})\cdot\sigma(X_{h,k})}$$

r_j affects C_{Z_j} and imposes a dependence inside columns and so the parameters of Z_j must depend on r_j, through C_{Z_j}:

$$\mathbb{E}(Z_{i,j}) = \hat{m}_{i,j}, \qquad \sigma(Z_{i,j}) = \hat{C}_{Z_j}\cdot\hat{m}_{i,j} = C_{Z_j}\cdot r_j\cdot\hat{m}_{i,j}$$

4.6 One-Year Time Horizon

We will focus now on Reserve Risk for 1 Line of Business, in a 1 Year Time horizon.
The reserve should cover:

- **Payments occurred in** t+1 for existing claims, no new business;
- **The new claims Reserve estimated at time** t+1 for existing claims, no new business.

One of the most important variables to describe this kind of risks is the **Claim Development Result** of a single AY i:

$$CDR_i = R_i^{D^{(n)}} - X_{i,n-i+1} - R_i^{D^{(n+1)}} = C_{i,n}^{D^{(n)}} - C_{i,n}^{D^{(n+1)}}$$

In this formula we can see two different behaviour, the fact that the reserve estimated in year t, using information up to year t, should be enough to cover payments in the next years and the reserve at the end of the next year, using all information available up to $t+1$, but also the fact that it is the difference between the ultimate cost computed using information at t and the ultimate cost computed using information up to year $t+1$.

$$CDR = \sum_{i=0}^{n} CDR_i$$

when CDR is negative, a loss will be generated, otherwise a Gain is observed. Volatility for X is the same as in the total run-off case, while for R we only have the estimate, but we have also other value in $R^{D^{(n+1)}}$.

4.6.1 Merz and Wutrich Model

Merz and Wutrich uses the same assumptions as Mack Formula, that are classical Chain Ladder as underlying method.
For $i > 1$:

$$\begin{aligned}\mathbb{E}(CDR_i|D^{(n)}) &= \mathbb{E}\left(\sum_{j=n-i+1}^{n} X_{i,j}|D^{(n)}\right) - \mathbb{E}\left(X_{i,n-i}|D^{(n)}\right) - \mathbb{E}\left(\mathbb{E}\left(\sum_{j=n-i+2}^{n} X_{i,j}|D^{(n+1)}\right)|D^{(n)}\right) = \\ &= \mathbb{E}\left(\sum_{j=n-i+2}^{n} X_{i,j}|D^{(n)}\right) - \mathbb{E}\left(\mathbb{E}\left(\sum_{j=n-i+2}^{n} X_{i,j}|D^{(n+1)}\right)|D^{(n)}\right)\end{aligned}$$

Under Classical Chain Ladder:

$$\hat{\lambda}_j^{D^{(n)}} = \frac{\sum_{i=0}^{n-j-1} C_{i,j+1}}{\sum_{i=0}^{n-j-1} C_{i,j}} \qquad \forall\ j = 0, 1, ..., n-1$$

$$\hat{\lambda}_j^{D^{(n+1)}} = \frac{\sum_{i=0}^{n-j} C_{i,j+1}}{\sum_{i=0}^{n-j} C_{i,j}} \qquad \forall\ j = 1, 2, ..., n-1$$

$$\mathbb{E}\left(\hat{\lambda}_j^{D^{(n+1)}}|D^{(n)}\right) = \mathbb{E}\left[\frac{\sum_{i=0}^{n-j} C_{i,j+1}}{\sum_{i=0}^{n-j} C_{i,j}}|D^{(n)}\right] = \frac{\sum_{i=0}^{n-j-1} C_{i,j+1} + \mathbb{E}(C_{n-j,j+1}|D^{(n)})}{\sum_{i=0}^{n-j} C_{i,j}} =$$

$$= \frac{\hat{\lambda}_j^{D^{(n)}} \cdot \sum_{i=0}^{n-j-1} C_{i,j} + \hat{\lambda}_j^{D^{(n)}} \cdot C_{n-j,j}}{\sum_{i=0}^{n-j} C_{i,j}} = \hat{\lambda}_j^{D^{(n)}}$$

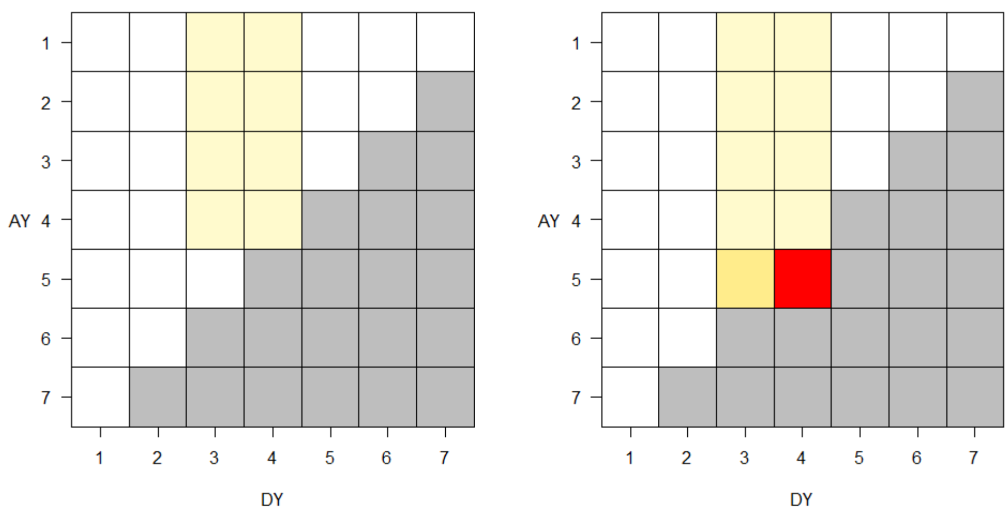

Figure 4.1: An example for one triangle 7 by 7, where in left one we have the cells used for estimating $\hat{\lambda}_j^{D^{(n)}}$, while on the right triangle are highlighted the cells involved in the estimation of $\hat{\lambda}_j^{D^{(n+1)}}$

As we can see from the picture 4.1, the estimate for $\hat{\lambda}_j^{D^{(n)}}$ is given by the values in light yellow in the left triangle, which are known given the initial triangle $D^{(n)}$. The estimate for $\hat{\lambda}_j^{D^{(n+1)}}$, in the right triangle, is made by using the same light yellow cells, but also the orange one corresponding generally to $C_{n-j,j}$, which is known, and the red cell, which is $C_{n-j,j+1}$, which is not known given triangle $D^{(n)}$, but needs to be estimated, in Expected Value we have that $\mathbb{E}(C_{n-j,j+1}|D^{(n)}) = C_{n-j,j} \cdot \hat{\lambda}_j^{D^{(n)}}$. For this reason the estimation of λ_j in both

triangle is the same, given the fact that we estimated the red cell with $\hat{\lambda}_j^{D^{(n)}}$.
For this reason we can state that the **Expected Value of CDR_i** is Zero.

$$\begin{aligned}\mathbb{E}(CDR_i|D^{(n)}) = \mathbb{E}(C_{i,n}^{D^{(n)}} - C_{i,n}^{D^{(n+1)}}|D^{(n)}) = \\ &= \prod_{j=n-i}^{n-1} \mathbb{E}(\hat{\lambda}_j^{D^{(n)}}|D^{(n)}) \cdot C_{i,n-i} - \prod_{j=n-i}^{n-i} \mathbb{E}(\hat{\lambda}_j^{D^{(n+1)}}|D^{(n)}) \cdot C_{i,n-i} = \\ &= \prod_{j=n-i}^{n-1} \hat{\lambda}_j^{D^{(n)}} \cdot C_{i,n-i} - \prod_{j=n-i}^{n-i} \hat{\lambda}_j^{D^{(n)}} \cdot C_{i,n-i} = 0\end{aligned}$$

This result holds for example in Local GAAP, while for solvency 2, we have other factors to take into account, such as discounting, which will lead to a different value of the CDR_i, often negative if the interest rates are positive. Make attention to the fact that in Solvency 2 we have a negative CDR does not mean that we would incur in a loss, because we have to take into account also the positive effect of the return on assets covering the reserve between t and $t+1$, which would act as a compensation.
Another important thing to take into account is the effect of the Risk Margin because in Solvency II, we would have that the total CDR is:

$$CDR_i = \left(BE^{D^{(n)}} + RM^{D^{(n)}}\right) - X - \left(BE^{D^{(n+1)}} + RM^{D^{(n+1)}}\right)$$

So also the difference between the two RM should be taken into account, called **Risk Margin Release**:

$$\mathbb{E}\left(RM^{D^{(n)}} - RM^{D^{(n+1)}}\right) > 0$$

This is mainly due to the fact that, if we take into account a closed portfolio, each year the RM will decrease, due to the decrease in the remaining Reserve.
For capital requirement purposes, usually Risk Margin is not considered, because risk margin depends on the capital requirements, and this would create a looping mechanism difficult to take into account in closed formulas.
Prediction Variance of the Claims Development Result will depend upon:

- Process Variance of X in a 1 year view;
- Estimation Variance of X in a 1 year view;
- Estimation Variance of $R^{D^{(n+1)}}$

$$msep\left(C\hat{D}R_i|D^{(n)}\right) = \mathbb{E}\left[\left(C\hat{D}R_i - 0\right)^2 |D^{(n)}\right] = \mathbb{E}\left[\left(\hat{C}_{i,n}^{D^{(n)}} - \hat{C}_{i,n}^{D^{(n+1)}}\right)^2 |D^{(n)}\right] =$$

$$= \hat{C}_{i,n}^2 \cdot \left[\underbrace{\frac{\hat{\sigma}_{n-i}^2}{\hat{\lambda}_{n-i}^2 \cdot C_{i,n-i}}}_{\text{PROC. VAR. for } X} + \underbrace{\frac{\hat{\sigma}_{n-i}^2}{\hat{\lambda}_{n-i}^2 \cdot \sum_{h=0}^{i-1} C_{h,n-i}}}_{\text{EST. VAR. for X}} + \underbrace{\sum_{j=n-i+1}^{n-1} \boxed{\frac{C_{n-j,j}}{\sum_{i=0}^{n-j} C_{i,j}}} \cdot \left(\frac{\hat{\sigma}_j^2}{\hat{\lambda}_j \cdot \sum_{h=0}^{n-j-1} C_{h,j}} \right)}_{\textbf{EST. VAR. for } R_i^{D^{(n+1)}}} \right]$$

The boxed term in the Estimation Variance for $R_i^{D^{(n+1)}}$ is in the range $(0, 1)$, and it scales the estimation variance, which is basically due to the reduction of parameter uncertainty due to the new information available.
Estimates of λ_j and σ_j^2 are the same defined in Mack Formula.

$$msep(CD\hat{R)}|D^{(n)}) = \sum_{i=1}^{n} msep(C\hat{D}R_i|D^{(n)}) +$$

$$+ 2 \cdot \sum_{i=1}^{n-1} \hat{C}_{i,n} \cdot \sum_{h=i+1}^{n} \hat{C}_{h,n} \cdot \left[\frac{\hat{\sigma}_{n-i}^2}{\hat{\lambda}_{n-i}^2 \cdot \sum_{h=0}^{i-1} C_{h,n-i}} + \sum_{j=n-i+1}^{n-1} \frac{C_{n-j,j}}{\sum_{i=0}^{n-j} C_{i,j}} \cdot \left(\frac{\hat{\sigma}_j^2}{\hat{\lambda}_j^2 \cdot \sum_{h=0}^{n-j-1} C_{h,j}} \right) \right]$$

As for Mack Formula covariance Between two different AY CDR can easily be derived:

$$Cov(C\hat{D}R_i, C\hat{D}R_h) = \frac{\hat{C}_{h,n}}{\hat{C}_{i,n}} \cdot Var(C\hat{D}R_i)$$

Merz And Wuthrich can be used in:

- STANDARD FORMULA: USP RESERVE RISK, using the parameters estimated with MW instead of Market Wide coefficients, in a weighted sum, with weights equal to $c_L oB$, depending on the length of the time series at disposal:

$$\sigma_{USP} = \sigma_{market} \cdot (1 - c_{LoB}) + \sigma_{Merz} \cdot c_{LoB};$$

- INTERNAL MODEL: we make assumptions about the distribution of the risk, and then we use the estimate of variance, using Merz and Wuthrich, and expected value in order to get the desired percentile (99.5% for Solvency II). Usually the distribution is calibrated by using the method of moments and the estimators obtained by the underlying method

4.6.2 Stochastic Re-Reserving

Stochastic Re-Reserving is a methodology based on an underlying Bootstrap-Simulation Technique used in order to derive the distribution of the Reserve in a One Year Time Horizon.

We assume that the Total Run-Off is based on a Bootstrap + Simulation (ODP) Model.

For $X_{i,j}$ the volatility in the total run-off is the same as for 1 Year Time Horizon, both for process and estimation. And so we get for each cell of $X_{i,j}$ the distribution of the expected value.

For what concerns the computation of $EV(R)$, we need to build $R^{D^{(n+1)}}$; Given the triangles $\ddot{P}_{i,j}^{h}$, for $h = 1, ..., H$, we take the values in the first diagonal $\ddot{P}_{i,j}^{h}$ with $i + j = n + 1$, and replace the values in the upper triangle with the real values of incremental payments: $P_{i,j}$. Now we have a triangle containing the information of the initial triangle, of the simulation of each step h of the first lower triangle, and we need to estimate with the used methodology the remaining lower cells (we used Paid Chain Ladder), once we have fitted our model, we now build the reserve for each AY i, and the total reserve. At the end we will have two different distributions, the distributions of $X_{i,j}$ in the first lower diagonal, and the distribution of the new computed reserve $R^{D^{(n+1)}}$, from which we can get the expected value and the variance, but also skewness and kurtosis, and other useful indicators.

This methodology takes implicitly into account the dependency between X and R.

We can include the Discounting, not present in usual Run-Off Bootstrap + Simulation, in order to compute the Solvency II Capital Requirement (SCR), while in closed formulas is more complicated. The assumption used in closed formula is that the coefficient of volatility remains the same for both discounted and un-discounted reserve.

One of the main results is that although the CoV is different for different methodologies (Merz and Wuthrich and Bootstrapping), the ratio between CoV in a One Year Framework, and the CoV in a Total Run-Off Framework is pretty stable.

The fact that different methodologies lead to different results of CoV, implies that also the SCR computed are different, Supervisor Authorities pay a great attention to the statistical significance of the underlying assumptions of the methods used, making sure that the methodology chose is the one that best fits the data, and not the one leading to the lower value of SCR. There is a problem in testing the significance of the assumptions for the data, given the fact that most part of the test used rely on small quantity of data, becoming sort of meaningless, but for example the distribution chosen in the bootstrap methodology, can be tested given the fact that single claims data is used.

4.7 One Year Approach and Capital Requirement

Knowing the characteristics of the r.v. CDR, it is possible to compute the SCR for reserve risk (related to outstanding loss liabilities of past exposure claims). To this aim, we define the r.v. Y:

$$Y = \sum_{i=1}^{n} X_{i,n-i+1} + R^{D^{(n)}}$$

we can see that if $\mathbb{E}(CDR_i) = 0$ then $\mathbb{E}(Y) = R^{D^{(n)}}$, $\sigma(CDR) = \sigma(Y)$, and $\gamma_1(CDR) = -\gamma_1(Y)$.
And so the SCR at confidence level α is defined as:

$$SCR_{\alpha}^{res} = q_{\alpha}(Y) - R^{D^{(n)}}$$

Where $q_{\alpha}(Y)$ is the percentile at level α of the r.v. Y. As can be seen in the Figure 4.2:

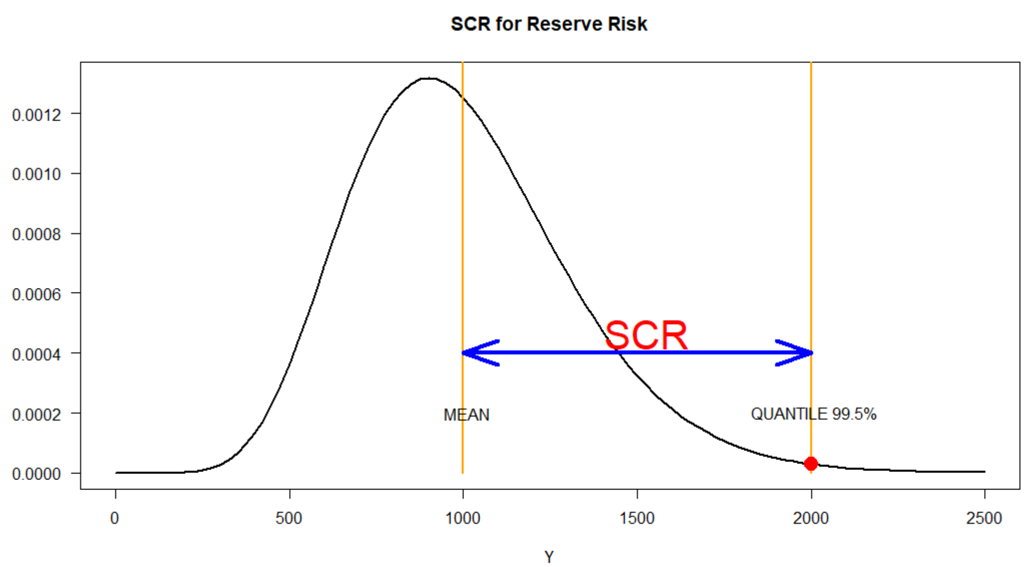

Figure 4.2: SCR for Reserve Risk

Notice that:

- SCR can also be directly computed by using the distribution of CDR;

- If we use discounted Reserves, we should also take into account of allocated investment income on the reserve hels;

- If we use market consisten value of Reserve, we need to consider the Risk Margin both for R^{D^n} and $R^{D^{n+1}}$

In Solvency II regulation, Reserve Risk SCR is computed jointly with Premium Risk, but if we consider only Reserve Risk, **Standard Formula** defines the following capital Requirement:

$$SCR_{99.5\%}^{res,SF} = 3 \cdot \sigma \cdot V$$

where V is the volume measure, equal to the net best estimate, and σ is a relative volatility measure that cathces the volatility of the ratio $\dfrac{CDR}{BE^{D^{(n)}}}$
In the Standard formula, we do not consider Risk Margin, neither Safety Loading in premium Risk.
The formula is based on a multiplier (in this case 3), which is based on the assumption that the underlying distribution is a lognormal with CoV equal to around 15%, this assumption is used in order to simplify the formula, but in reality the real multiplier should depend on the real CoV of the CDR, if the distribution chosen is lognormal.
So in Standard Formula we are overestimating the SCR if the real CoV estimated is smaller then 15%, and underestimating SCR when CoV is higher.
$\sigma \approx \frac{\sigma(Y)}{BE^{D^{(n)}}}$, in market wide the coefficient σ is given by a table, in USP (Undertaking Specific Parameter) we have to possible methods which are given by the calculation of the run-off ratio volatility or one based on Merz and Wutrich, in both cases the final value to be used in the standard formula is given by a weighted average of market wide and USP estimates, using two credibility factors, depending on the LoB: $\sigma = (1 - c_{LoB}) \cdot \sigma_{MKT} + c_{LoB} \cdot \sigma_{USP}$

4.7.1 SCR Aggregation

Once we have an SCR for all our different LoBs, we need to calculate the total SCR for portfolio, this can be done in different ways:

- Linear Correlation Aggregation: Is the methodology used in the Standard Formula. The final SCR is given by the same formula $3 \cdot \sigma_{portfolio} \cdot V$ where $\sigma_{portfolio}$ is compute using the Variance/Covariance Matrix for aggregating the different σ_{LoB}. This is easy to do, but it does not catch the possible tail dependency between different LoBs, but also other type of non-linear dependencies.

- Through the use of Copula Functions: Copulas are functions to take pass from the marginal distribution of single risks, to the joint distribution of the total risk.

Chapter 5

Rate-making in MTPL Line of Business

5.1 Risk Premium

As well-known, in rate-making of non-life insurance business, the first step is the assessment of expected aggregate claim amount of claims at portfolio level as:

$$P = \mathbb{E}(X)$$

where the random variable X (aggregated claim amount) considers the claims (paid and reserved) in period in which the insurance company will sell the policies. This situation lead to 2 important aspects:

- the r.v. X is related to the future behavior of the policies (contracts sell in 2023). This is because, since we are interested in the value of premium, the payment for claims occurred before than 2023 are covered by the claim reserve;
- the r.v. X considers not only payment of 2023 but also the reserve, so we are estimating a r.v. that consider an estimation (the value of the reserve of 2022). For this reason we are conditioning the expected value of X to the uncertainty of R.

Usually the estimate of the average fair premium is based on separate valuation of average cost (AC) and frequency (f) of different accident years:

$$\overline{P} = \mathbb{E}(AC) \cdot \mathbb{E}(f)$$

5.2 Estimation of "basic" Average Cost and Frequency

The estimation of the Average cost and expected frequency is based on the empirical analysis of the portfolio over the last year: the regulation requires the insurer at least a time-series of the last 5 years. Obviously the undertaking can decide to partially include (or not include at all) some years where the behavior of frequency and AC is not consistent with the expectation (pandemic is an example where the frequency decreased a lot due to a minor urban traffic caused by lockdown).
In order to compute the premium the insurer only considers:

- the number of open claims (paid and reserve). Reported claims are not considered since they can cause some problem of valuation (claims closed w/o payment, re-open claims),
- the exposure of the time span in which the contract has been underwritten (if a policy is sold in 01/06/2022 the exposure for the year is 0.5)

Then we have that the **frequency** is given by: number of claims over the exposure; the **AC** is obtained as total amount of claims (paid and reserved) over the number of open claims (paid and reserved).
At the end we got a number of values equal to the number of year of the time-series used (if the company uses exactly 5 years time-series then it will got 5 values for the frequency and 5 value for the AC).
AC and frequency have (by construction) some uncertainty:

- reserved amount: as always it is an estimation so it can be lower than the real payment (insufficient reserve) so we compute future premium with a value that is not able to cover the payment
- number of claims: re-open and closed w/o payment affect the total amount of claims
- IBNR: only reported claims considered since the IBNR will for sure increase the value of the frequency, but we do not know in advance the effect on the AC (it could increase or decrease).

5.2.1 Forecasting Final Average Cost (AC)

Basic AC can be adjusted to consider several aspects:

- future development of claims, considering: Reserve, IBNR, closure and re-opening
- period of coverage of new premiums: data used are in t but the contract will be sold in t+2

- Settlement expenses: the insurer can have some expenses (i.e. legal) to prove that the policyholder has not the right to receive the money for the claim

Typically the cumulative coefficient are computed such that:

$$\widehat{AC} = \widehat{AC}_t \cdot \prod_{i=1}^{3} c_i^{AC} \quad \forall \quad i = 1, 2, 3$$

In order to consider future development of open claims, we need to estimate the proportion of future re-open claims and closed claims w/o payment. This amount can be positive or negative since the impact of these claims can increase or decrease the total amount of claims in the insurance portfolio. By using this percentage, we can estimate the new proportion of paid and open claims as:

$$\hat{p}^R = \frac{\frac{n^R}{n^R + n^P} + \hat{p}_{re}}{1 + \hat{p}_{re}}$$

$$\hat{p}^P = 1 - \hat{p}^R$$

where n^P and n^R are number of paid and reserved claims. Pay attention: when considering the $\hat{p}_{re}$, at denominator we will have an higher amount of claims if the re-open are higher than closed (and a lower denominator if the vice-versa occurs).

coefficient c1

We need to estimate the AC_t^R in order to take into account the sufficiency/insufficiency effect of the reserve (the AC_t^P is already at our disposal). Pay attention: since at denominator we have the $\widehat{p}^R$ (that is an amount that does not reaches 100 per cent), we need to normalize the settlement speed (as we do also in other procedure, i.e. Fisher-Lange).

$$\hat{AC}_t^R = \frac{\hat{C}_t}{\hat{p}^R} = \frac{\sum_{i=1}^{T} C_{t,i}}{\hat{p}^R} = \frac{\sum_{i=1}^{T} \hat{AC}_{t,i} \cdot s_i^{N,I}}{\hat{p}^R}$$

$s_i^{N,I} = s_i^I \cdot \frac{\hat{p}^R}{1 - s_0^C}$ is the normalized incremental settlement speed, that is the incremental settlement speed s_i^I divided by the sum of the incremental settlement speeds from 1 to T: $1 - s_0^C$, all multiplied by the factor $\hat{p}^R$, in order to have that $\sum_{i=0}^{T} s_i^{N,I} = \hat{p}^R$.

$\hat{C}_{t,i} = \hat{AC}_{t,i} \cdot s_i^{N,I}$ is the average cost of open claims multiplied by the Incremental Normalized settlement speed (Attention: $\hat{C}_{t,i}$ is not a cumulative payment). We can see how the **Average Cost of Open Claims** is the **weighted average** of the $C_{t,i}$, weighted by $s_i^{I,N}$.

The factor $c_r = \frac{\hat{AC}^R}{AC^R}$ gives us a proportion of the impact of the new estimate of the Average cost of open claims with respect to the older.

$$\hat{c}_1^{AC} = \frac{\hat{p}^P \cdot AC_t^P + \hat{p}^R \cdot \hat{AC}_t^R}{\hat{AC}_t}$$

coefficient 2

We need to introduce a correction in order to take into account the **period of coverage**. There is a time lag between data at out disposal (up to t, so 2022 in our case) and period in which the company will sell the policies (t+1). So this coefficient considers a correction both for AC and frequency:
A common change in the cost can be the presence of different inflation rate for the future (so the company expects to pay more in future because of an increase in the inflation), or simply a decreasing behaviour of the frequency (for example the decrease of traffic in different cities for the pandemic).
Coverage period considers:

- beginning of coverage period of first contract sold
- end of coverage period of last contract sold

If the first contract is sold on 01/01/2023 (assume only annual contracts) and the last is sold on 31/12/2023, our coverage period will end on 31/12/2024 (so given we are in 2022 nowadays, we will stop in t+2). Then we want to see also the proportion of when the contracts are sold: uniform distribution (so we assume they are sold all in June) or a different distribution (so month with an higher amount of contracts sold)?
Obviously, an important aspect when dealing with insurance business is the portion of contracts will generate claims in future ϕ_{t+h}, for example, if annual policies are sold from the first of July t+2 to the end of June t+3 (and we assume uniform distribution of premiums and claims), we have: $\phi_{t+2} = 0.25$, $\phi_{t+h3} = 0.5$, $\phi_{t+4} = 0.25$.

- $\hat{\phi}_{t+h}$ is the Proportion of Claims incurred in each calendar year covered, with $\sum_h \hat{\phi}_{t+h}$
- $\hat{AC}^P = \frac{\sum_h C_{t+h,0} \cdot \hat{\phi}_{t+h}}{\sum_h \hat{p}_{t+h}^P \cdot \hat{\phi}_{t+h}} = \frac{\sum_h \hat{AC}_{t+h,0} \cdot \hat{p}_{t+h}^P \cdot \hat{\phi}_{t+h}}{\sum_h \hat{p}_{t+h}^P \cdot \hat{\phi}_{t+h}}$;
- $\hat{p}_{t+h}^P$ is the estimate of the settlement speed in future years;

- $\hat{AC}^P = \frac{\sum_{j=1}^{T}\sum_h C_{t+h,j} \cdot \hat{\phi}_{t+h}}{\sum_h \hat{p}^R_{t+h} \cdot \hat{\phi}_{t+h}} = \frac{\sum_{j=1}^{T}\sum_h \hat{AC}_{t+h,j} \cdot \hat{s}^I_{t+h,j} \cdot \hat{\phi}_{t+h}}{\sum_h \hat{p}^R_{t+h} \cdot \hat{\phi}_{t+h}}$;

$$\hat{c}_2^{AC} = \frac{\hat{p}^{P*} \cdot \hat{AC}^P + \hat{p}^{R*} \cdot \hat{AC}^R}{\hat{p}^P \cdot AC_t^P + \hat{p}^R \cdot \hat{AC}_t^R}$$

where $\hat{p}^{P*} = \sum_h \hat{p}^P_{t+h} \cdot \hat{\phi}_{t+h}$, and $\hat{p}^{R*} = \sum_h \hat{p}^R_{t+h} \cdot \hat{\phi}_{t+h}$

coefficient 3

Is the handling expense, usually it is computed by accounting, and added to the premium computation.

Final Average Cost

Once we have estimated the three coefficients, c_1^{AC}, c_2^{AC}, c_3^{AC}, we can compute the final average cost by:

$$\overset{\star}{AC}_t = AC_t \cdot c_1^{AC} \cdot c_2^{AC} \cdot c_3^{AC} \cdot c_{fin}^{AC}$$

where c_{fin}^{AC} is the discount coming from the investment of the premiums from collection to payment, usually lower then 1.

5.2.2 Frequency

We start from the basic frequency estimated from data, f_t, we need to estimate:

- c_1^f is the correction for IBNR and re-opened claims;
- c_2^f effect of the coverage period (from t to t+2)

$$\overset{\star}{f}_t = f_t \cdot c_1^f \cdot c_2^f$$

5.2.3 Average Pure Premium

Is the product between the final frequency and the final average cost:

$$P_t = \overset{\star}{f}_t \cdot \overset{\star}{AC}_t$$

5.2.4 Average Gross Premium

The items to be added to the pure premiums are:

- Safety Loadings: λ
- Expense Loadings: c^{EXP}
- contibutions to FGVS (Fondo di Garanzia Vittime della Strada): c^{FGVS}

$$B_t = P_t \cdot \left(c^{FGVS} + \frac{1+\lambda}{1-c^{EXP}} \right)$$

5.2.5 Specific Premium Rate

In the insurance market, the premium rates are personalized based on different risk factors, so that different policyholders will have to pay different risks (and not everyone paying the average premium) everyone depending on their risks. If we consider for example 3 different risk factors we will have:

$$P_{i,j,h} = P_b \cdot \alpha_i \cdot \beta_j \cdot \gamma_h$$

One important thing is that insurance companies will need to have an equilibrium given by the average premium multiplied by the number of policies sold:

$$\sum_{v=1}^{V} P_v^{spec} = V \cdot P_t$$

where V is the number of policies sold, P_v^{spec} is the specific premium for the v-th policyholder. One way to compute the Basic Premium is

$$P_b = \frac{P \cdot V}{\sum_{i,j,h} P_b \cdot \alpha_i \cdot \beta_j \cdot \gamma_h \cdot v_{i,j,h}} \approx \frac{P}{\overline{\alpha} \cdot \overline{\beta} \cdot \overline{\gamma}}$$

$v_{i,j,h}$ is the number of policies for risk factor 1 equal to mode i, risk factor 2 equal to mode j and risk factor 3 equal to mode h, with $\sum_{i,j,h} v_{i,j,h} = V$.

Customization of the premium is a very important thing for the commercial point of view and it is important that the analysis made before the policy is sold are strongly accurate.

Chapter 6

Reinsurance Treaties

Chapter 7

Appendix

7.1 Proof 1: Process Variance

$$\begin{aligned}
&\boxed{Var(C_{i,n}|D^{(n)})} = Var(C_{i,n}|C_{i,n-i}) = \\
&= \mathbb{E}\left[Var(C_{i,n}|C_{i,n-i})|C_{i,n-i}\right] + Var\left[\mathbb{E}(C_{i,n}|C_{i,n-i})|C_{i,n-i}\right] \\
&= \sigma^2_{n-1} \cdot \mathbb{E}(C_{i,n}|C_{i,n-i}) + Var(\lambda_{n-1} \cdot C_{i,n-1}|C_{i,n-i}) \\
&= \sigma^2_{n-1} \cdot \prod_{h=n-i}^{n-1} \lambda_h \cdot C_{i,n-i} + \lambda^2_{n-1} \cdot \boxed{Var(C_{i,n-1|C_{i,n-i}})}
\end{aligned}$$

As we can see this formula is recursive, meaning that the term at step n depends on the term at step $n-1$. In order to get to the closed formula we give an example:
For $i = 3$:

$$\begin{aligned}
&Var(C_{3,n}|C_{3,n-3}) = \sigma^2_{n-1} \cdot \lambda_{n-2} \cdot \lambda_{n-3} \cdot C_{3,n-3} + \lambda^2_{n-1} \cdot Var(C_{3,n-1}|C_{3,n-3}) \\
&Var(C_{3,n-1}|C_{3,n-3}) = \sigma^2_{n-2} \cdot \lambda_{n-3} \cdot C_{3,n-3} + \lambda^2_{n-2} \cdot Var(C_{3,n-2}|C_{3,n-3}) \\
&Var(C_{3,n-2}|C_{3,n-3}) \underbrace{=}_{def.} C_{3,n-3} \cdot \sigma^2_{n-3}
\end{aligned}$$

By applying the formulas recursively we obtain:

$$\begin{aligned}
Var(C_{3,n}|C_{3,n-3}) &= \sigma^2_{n-1} \cdot \lambda_{n-2} \cdot \lambda_{n-3} \cdot \underline{C_{3,n-3}} + \lambda^2_{n-1} \cdot \left[\sigma^2_{n-2} \cdot \lambda_{n-3} \cdot \underline{C_{3,n-3}} + \lambda^2_{n-2} \cdot \left(\underline{C_{3,n-3}} \cdot \sigma^2_{n-3}\right)\right] \\
&= C_{3,n-3} \cdot \left(\underline{\sigma^2_{n-1}} \cdot \lambda_{n-3} \cdot \lambda_{n-2} + \lambda^2_{n-1} \cdot \underline{\sigma^2_{n-2}} \cdot \lambda_{n-3} + \lambda^2_{n-1} \cdot \lambda^2_{n-2} \cdot \underline{\sigma^2_{n-3}}\right) = \\
&= C_{3,n-3} \cdot \sum_{m=n-3}^{n-1} \sigma^2_m \cdot \prod_{h=n-3}^{m-1} \lambda_h \cdot \prod_{k=m+1}^{n-1} \lambda_k
\end{aligned}$$

Which was what we wanted do demonstrate, know it is easy to show how to generalize from $i = 3$ to a generic i.

7.2 Proof 2: Estimation Variance

By using the following relationship:

$$s_k = \prod_{t=n-i}^{k-1} \widehat{\lambda}_t \cdot (\lambda_k - \widehat{\lambda}_k) \cdot \prod_{t=k+1}^{n-1} \lambda_t$$

we can prove that

$$\left(\prod_{t=n-i}^{n-1} \lambda_t - \prod_{t=n-i}^{n-1} \widehat{\lambda}_t \right)^2 = \left(\sum_{k=n-i}^{n-1} s_k \right)^2$$

imagine to have a 2 by 2 triangle and $i = 2$, $n = 2$:

$$\sum_{k=0}^{1} s_k = s_0 + s_1 = (\lambda_0 - \widehat{\lambda}_0) \cdot \lambda_1 + \widehat{\lambda}_0 \cdot (\lambda_1 - \widehat{\lambda}_1) =$$

$$= \lambda_0 \cdot \lambda_1 - \cancel{\widehat{\lambda}_0 \cdot \lambda_1} + \cancel{\widehat{\lambda}_0 \cdot \lambda_1} - \widehat{\lambda}_0 \cdot \widehat{\lambda}_1 =$$

$$= \prod_{t=0}^{1} \lambda_t - \prod_{t=0}^{1} \widehat{\lambda}_t$$

We can easily generalize the latter idea to generic values of i and n, as we wanted to proof.

The estimation variance is defined as:

$$\left(\hat{C}_{i,n} - \mathbb{E}(C_{i,n} | D^{(n)}) \right)^2 = C_{i,n-i}^2 \cdot \left(\prod_{t=n-i}^{n-1} \widehat{\lambda}_t - \prod_{k=n-i}^{n-1} \lambda_k \right)^2 = C_{i,n-i}^2 \cdot \left(\sum_{k=n-i}^{n-1} S_k \right)^2 =$$

$$= \underbrace{C_{i,n-1}^2 \cdot \sum_{k=n-i}^{n-1} s_k^2}_{\text{First Term}} + \underbrace{C_{i,n-1}^2 \cdot 2 \cdot \sum_{n-i \le h < k \le n-1} s_k \cdot s_h}_{\text{Second Term}}$$

We will focus firstly on the covariance terms, but first we need to define the set $B^{(k)} = \{C_{i,j} : i + j \le n \ \ and \ \ j \le k\} \subset D^{(n)}$:

$$\mathbb{E} \left(\sum_{n-i \le h < k \le n-1} s_k \cdot s_h | B^{(k)} \right) = \sum_{n-i \le h < k \le n-1} s_h \cdot \mathbb{E}(s_k | B^{(k)}) =$$

$$= \sum_{n-i \le h < k \le n-1} s_h \cdot \prod_{t=n-i}^{k-1} \widehat{\lambda}_t \cdot \mathbb{E} \left((\lambda_k - \widehat{\lambda}_k) | B^{(k)} \right) \cdot \prod_{t=k+1}^{n-1} \lambda_t = 0$$

Which was made by using the definition of s_k, the term result as Zero, because the expectation of $\lambda_k - \widehat{\lambda}_k$ conditional on $D^{(n)}$, and so also on $B^{(k)}$ is Zero.
So we have that the only term remaining is the first one.

$$\left(\hat{C}_{i,n} - \mathbb{E}(C_{i,n} | D^{(n)}) \right)^2 = C_{i,n-1}^2 \cdot \mathbb{E} \left(\sum_{k=n-i}^{n-1} s_k^2 | B^{(k)} \right) \overbrace{=}^{Linearity} C_{i,n-1}^2 \cdot \sum_{k=n-i}^{n-1} \boxed{\mathbb{E} \left(s_k^2 | B^{(k)} \right)}$$

We focus now on the highlighted term

$$\mathbb{E}\left(s_k^2|B^{(k)}\right) \overbrace{=}^{Definition} \mathbb{E}\left(\prod_{t=n-i}^{k-1}\widehat{\lambda}_t^2\cdot(\lambda_k-\widehat{\lambda}_k)^2\cdot\prod_{t=k+1}^{n-1}\lambda_t^2|B^{(k)}\right)=$$

$$=\prod_{t=n-i}^{k-1}\widehat{\lambda}_t^2\cdot\mathbb{E}\left((\lambda_k-\widehat{\lambda}_k)^2|B^{(k)}\right)\cdot\prod_{t=k+1}^{n-1}\lambda_t^2=$$

$$=\prod_{t=n-i}^{k-1}\widehat{\lambda}_t^2\cdot\boxed{\boxed{Var\left(\widehat{\lambda}_k|B^{(k)}\right)}}\cdot\prod_{t=k+1}^{n-1}\lambda_t^2$$

We now focus another time on the highlighted cell:

$$Var\left(\widehat{\lambda}_k|B^{(k)}\right) \overbrace{=}^{\text{Paid Chain Ladder}} Var\left[\frac{\sum_{i=0}^{n-k-1}C_{i,k+1}}{\sum_{i=0}^{n-k-1}C_{i,k}}|B^{(k)}\right]=$$

$$=\frac{Var\left[\sum_{i=0}^{n-k-1}C_{i,k+1}|B^{(k)}\right]}{\left(\sum_{i=0}^{n-k-1}C_{i,k}\right)\cdot\left(\sum_{i=0}^{n-k-1}C_{i,k}\right)}=\frac{\sum_{i=0}^{n-k-1}Var\left[C_{i,k+1}|B^{(k)}\right]}{\left(\sum_{i=0}^{n-k-1}C_{i,k}\right)\cdot\left(\sum_{i=0}^{n-k-1}C_{i,k}\right)}=$$

$$=\frac{\sum_{i=0}^{n-k-1}C_{i,k}\cdot\sigma_k^2}{\left(\sum_{i=0}^{n-k-1}C_{i,k}\right)\cdot\left(\sum_{i=0}^{n-k-1}C_{i,k}\right)}=\frac{\cancel{\left(\sum_{i=0}^{n-k-1}C_{i,k}\right)}\cdot\sigma_k^2}{\cancel{\left(\sum_{i=0}^{n-k-1}C_{i,k}\right)}\cdot\left(\sum_{i=0}^{n-k-1}C_{i,k}\right)}=$$

$$=\frac{\sigma_k^2}{\left(\sum_{i=0}^{n-k-1}C_{i,k}\right)}$$

So, by replacing the terms double boxed we get:

$$\mathbb{E}\left(s_k^2|B^{(k)}\right)=\prod_{t=n-i}^{k-1}\widehat{\lambda}_t^2\cdot\boxed{\boxed{\frac{\sigma_k^2}{\left(\sum_{i=0}^{n-k-1}C_{i,k}\right)}}}\cdot\prod_{t=k+1}^{n-1}\lambda_t^2$$

and then by replacing in the first boxed equation we get:

$$\left(\hat{C}_{i,n}-\mathbb{E}(C_{i,n}|D^{(n)})\right)^2=C_{i,n-1}^2\cdot\sum_{k=n-i}^{n-1}\boxed{\prod_{t=n-i}^{k-1}\widehat{\lambda}_t^2\cdot\boxed{\boxed{\frac{\sigma_k^2}{\left(\sum_{i=0}^{n-k-1}C_{i,k}\right)}}}\cdot\prod_{t=k+1}^{n-1}\lambda_t^2}$$

If we now expand the sum for k, we get:
for $(k = n-i)$:

$$C_{i,n-i}^2 \cdot \prod_{t=n-i}^{n-i-1} \widehat{\lambda}_t^2 \cdot \frac{\sigma_{n-i}^2}{\left(\sum_{l=0}^{i-1} C_{l,n-i}\right)} \cdot \prod_{t=n-i+1}^{n-1} \lambda_t^2 = C_{i,n-i}^2 \cdot 1 \cdot \frac{\sigma_{n-i}^2}{\left(\sum_{l=0}^{l-1} C_{l,n-i}\right)} \cdot \lambda_{n-i+1}^2 \cdot \ldots \cdot \lambda_{n-1}^2 \cdot \frac{\widehat{\lambda}_{n-i}^2}{\widehat{\lambda}_{n-i}^2}$$

for $(k = n-i+1)$:

$$C_{i,n-i}^2 \cdot \prod_{t=n-i}^{n-i} \widehat{\lambda}_t^2 \cdot \frac{\sigma_{n-i+1}^2}{\left(\sum_{l=0}^{i-2} C_{l,n-i+1}\right)} \cdot \prod_{t=n-i+2}^{n-1} \lambda_t^2 = C_{i,n-i}^2 \cdot \widehat{\lambda}_{n-i}^2 \cdot \frac{\sigma_{n-i+1}^2}{\left(\sum_{l=0}^{i-2} C_{i,n-i+1}\right)} \cdot \lambda_{n-i+2}^2 \cdot \ldots \cdot \lambda_{n-1}^2 \cdot \frac{\widehat{\lambda}_{n-i+1}^2}{\widehat{\lambda}_{n-i+1}^2}$$

And so on for every $k = n-i, n-i+1, ..., n-1$. The terms in red multiplied and divided don't change the terms, but they would be useful in order to complete the series of λ^2, in order to rewrite the series, together with $C_{i,n-i}^2$, as $\frac{C_{i,n}^2}{\widehat{\lambda}_k}$.
The only problem is the fact that each term is composed by both λ_t and $\widehat{\lambda}_t$.
When we estimate the Estimation variance we need to use the estimators of the real values, so λ becomes $\widehat{\lambda}$, and the same for $\widehat{\sigma}$ and $\widehat{C_{i,n}}$.
Hence the formula becomes:

$$\left(\hat{C}_{i,n} - \mathbb{E}(C_{i,n}|D^{(n)})\right)^2 = C_{i,n-1}^2 \cdot \sum_{k=n-i}^{n-1} \prod_{t=n-i}^{k-1} \widehat{\lambda}_t^2 \cdot \frac{\widehat{\sigma}_k^2}{\left(\sum_{i=0}^{n-k-1} C_{i,k}\right)} \cdot \prod_{t=k+1}^{n-1} \widehat{\lambda}_t^2 =$$

$$= \boxed{\hat{C}_{i,n}^2 \cdot \sum_{k=n-i}^{n-1} \left[\frac{\widehat{\sigma}_k^2}{\widehat{\lambda}_k^2 \cdot \sum_{i=0}^{n-k-1} C_{i,k}} \right]}$$

7.3 Proof 3: Variance of sum two generic cells

We assume to have tow cells, A and B, and we are interested in the **variance of the sum of the number of claims**, including the random variable $\tilde{q}$:

$$\begin{aligned}
Var\left(\tilde{K}_A + \tilde{K}_B\right) &= \mathbb{E}\left((\tilde{K}_A + \tilde{K}_B)^2\right) - \left(\mathbb{E}(\tilde{K}_A + \tilde{K}_B)\right)^2 = \\
&= \mathbb{E}\left(\tilde{K}_A^2 + \tilde{K}_B^2 + 2 \cdot \tilde{K}_A \cdot \tilde{K}_B\right) - (n_A + n_B)^2 = \\
&= \mathbb{E}(\tilde{K}_A^2) + \mathbb{E}(\tilde{K}_B^2) + 2 \cdot \mathbb{E}(\tilde{K}_A \cdot \tilde{K}_B) - (n_A + n_B)^2 = \\
&= \sigma^2(\tilde{K}_A) + \cancel{n_A^2} + \sigma^2(\tilde{K}_B) + \cancel{n_B^2} + 2 \cdot \mathbb{E}(\tilde{K}_A \cdot \tilde{K}_B) - (\cancel{n_A^2} + \cancel{n_B^2} + 2 \cdot n_A \cdot n_B) = \\
&= \sigma^2(\tilde{K}_A) + \sigma^2(\tilde{K}_B) + 2 \cdot \mathbb{E}(\tilde{K}_A \cdot \tilde{K}_B) - 2 \cdot n_A \cdot n_B
\end{aligned}$$

focusing now on the term $\mathbb{E}(\tilde{K}_A \cdot \tilde{K}_B)$:

$$\begin{aligned}
\mathbb{E}(\tilde{K}_A \cdot \tilde{K}_B) &= \mathbb{E}\left(\mathbb{E}(\tilde{K}_A \cdot \tilde{K}_B | \tilde{q} = q)\right) = \\
&= \mathbb{E}(n_A \cdot \tilde{q} \cdot n_B \cdot \tilde{q}) = \\
&= n_A \cdot n_B \cdot \mathbb{E}(\tilde{q}^2) = \\
&= n_A \cdot n_B \cdot \left(\sigma_{\tilde{q}}^2 + (\mathbb{E}(\tilde{q}))^2\right) = \\
&= n_A \cdot n_B \cdot \left(\sigma_{\tilde{q}}^2 + 1\right)
\end{aligned}$$

Now we replace the term $\mathbb{E}(\tilde{K}_A \cdot \tilde{K}_B)$ in the first series of equation:

$$\begin{aligned}
Var\left(\tilde{K}_A + \tilde{K}_B\right) &= \sigma^2(\tilde{K}_A) + \sigma^2(\tilde{K}_B) + 2 \cdot \left(n_A \cdot n_B \cdot \left(\sigma_{\tilde{q}}^2 + 1\right)\right) - 2 \cdot n_A \cdot n_B = \\
&= \sigma^2(\tilde{K}_A) + \sigma^2(\tilde{K}_B) + 2 \cdot n_A \cdot n_B \cdot \sigma_{\tilde{q}}^2 + \cancel{2 \cdot n_A \cdot n_B \cdot 1} - \cancel{2 \cdot n_A \cdot n_B} = \\
&= \sigma^2(\tilde{K}_A) + \sigma^2(\tilde{K}_B) + 2 \cdot n_A \cdot n_B \cdot \sigma_{\tilde{q}}^2
\end{aligned}$$

The correlation between K_A and K_B is given by:

$$\rho = \frac{2 \cdot n_A \cdot n_B \cdot \sigma_{\tilde{q}}^2}{\sigma(\tilde{K}_A) \cdot \sigma(\tilde{K}_B)}$$

So higher the term $\sigma_{\tilde{q}}^2$ the higher the covariance and so the correlation.